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**Re:** Supplement 1

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# Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants

API RECOMMENDED PRACTICE 941
FIFTH EDITION, JANUARY 1997
SUPPLEMENT 1, APRIL 1998



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# Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants

**Manufacturing, Distribution and Marketing Department**

API RECOMMENDED PRACTICE 941
FIFTH EDITION, JANUARY 1997
SUPPLEMENT 1, APRIL 1998



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# CONTENTS


| | | Page |
|---|---|---|
| 0 | INTRODUCTION | 1 |
| 1 | SCOPE | 1 |
| 2 | REFERENCES | 1 |
| | 2.1 Standards | 1 |
| | 2.2 Other References | 2 |
| 3 | OPERATING LIMITS | 3 |
| | 3.1 Bases For Setting Operating Limits | 3 |
| | 3.2 Selecting Materials For New Equipment | 3 |
| 4 | FORMS OF HIGH TEMPERATURE HYDROGEN ATTACK | 7 |
| | 4.1 General | 7 |
| | 4.2 Surface Decarburization | 7 |
| | 4.3 Internal Decarburization and Fissuring | 7 |
| 5 | FACTORS INFLUENCING HTHA | 8 |
| | 5.1 High Temperature Hydrogen Attack In A Liquid Hydrocarbon Phase | 8 |
| | 5.2 Incubation | 8 |
| | 5.3 Effect of Primary Stresses | 9 |
| | 5.4 Effect of Secondary Stresses | 10 |
| | 5.5 Effect of Heat Treatment | 10 |
| | 5.6 Effect of Stainless Steel Cladding or Weld Overlay | 10 |
| 6 | INSPECTION | 11 |
| | 6.1 Inspection For High Temperature Hydrogen Attack | 11 |
| APPENDIX A—HIGH TEMPERATURE HYDROGEN ATTACK OF 0.5Mo STEELS | | 13 |
| APPENDIX B—HIGH TEMPERATURE HYDROGEN ATTACK OF 1.25Cr–0.5Mo STEEL | | 21 |
| APPENDIX C—HIGH TEMPERATURE HYDROGEN ATTACK OF 2.25Cr–1Mo STEEL | | 23 |
| APPENDIX D—INSPECTION FOR HTHA TABLES | | 25 |
| APPENDIX E—REQUEST FOR NEW INFORMATION | | 31 |

Figures

| | |
|---|---|
| 1—Operating Limits for Steels in Hydrogen Service to Avoid Decarburization and Fissuring | 5 |
| 2—C–0.5Mo Steel Showing Internal Decarburization and Fissuring in High Temperature Hydrogen Service | 8 |
| 3—Time for Incipient Attack of Carbon Steel in High Temperature Hydrogen Service | 9 |
| A-1—Experience with C–0.5Mo and Mn–0.5Mo Steel | 15 |
| A-2—Steels in High Temperature Hydrogen Service Showing Effect of Molybdenum and Trace Alloying Elements | 18 |
| A-3—Time for Incipient Attack of 0.5Mo Steels in High Temperature Hydrogen Service | 19 |
| B-1—Operating Conditions for 1.25Cr–0.5Mo Steels That Experienced HTHA Below the Figure 1 Curve | 22 |


98

98

98



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Page

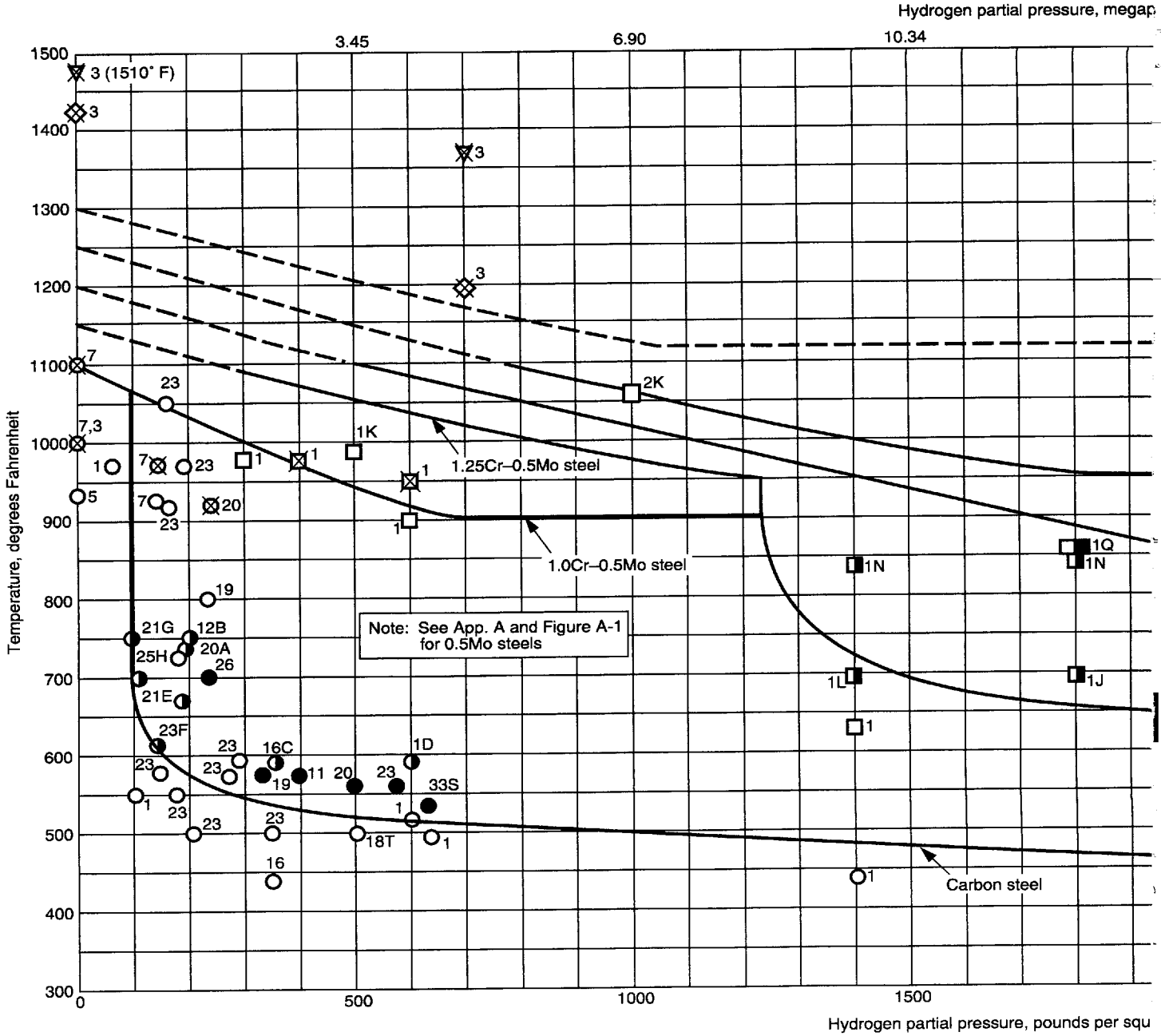
98 C-1—Operating Conditions of 2.25Cr–1Mo Steels That Experienced HTHA Below the Figure 1 Curve ..... 24

Tables

98 A-1—Operating Conditions for C–0.5Mo Steels That Experienced High Temperature Hydrogen Attack Below the 0.5Mo Steel Curve in Figure A-1 ..... 17
B-1—Experience with High Temperature Hydrogen Attack of 1.25Cr–0.5Mo Steel at Operating Conditions Below the Figure 1 Curve ..... 21
C-1—Experience with High Temperature Hydrogen Attack of 2.25Cr–1Mo Steel at Operating Conditions Below the Figure 1 Curve ..... 23
D-1—Summary of Inspection Methods for High Temperature Hydrogen Attack ..... 27
D-2—Summary of Inspection Methods for High Temperature Hydrogen Attack ..... 29



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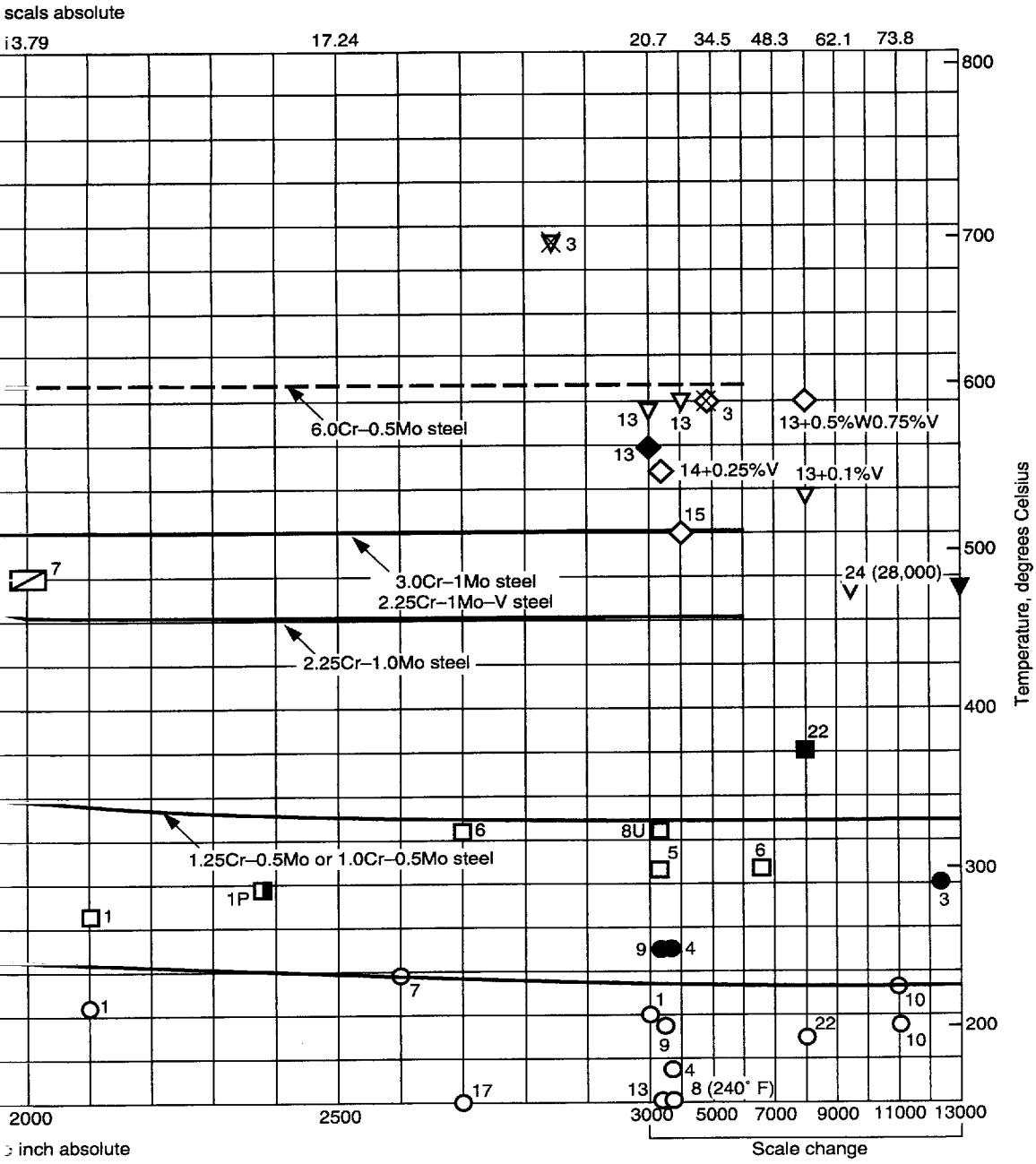
1. The limits described by these curves are based on service experience originally collected by G.A. Nelson and on additional information gathered by or made available to API.
2. Austenitic stainless steels are generally not decarburized in hydrogen at any temperature or hydrogen pressure.
3. The limits described by these curves are based on experience with cast steel as well as annelaed and normalized steels at stress levels defined by Section VIII, Division I, of the ASME code. See 5.3 and 5.4 in text for additional information.
4. Several failures of 1-1/4Cr–1Mo steel have been reported in the satisfactory region. See Appendix B for details.
5. The inclusion of the 2.25Cr–1Mo-V class of steels is based on 10,000+hr laboratory tests where these alloys were at least equal to the 3Cr–1Mo steel. See Reference 22 listed in Section 2.2.

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January 1997

Figure 1—Operating Limits for Steels in Hydrogen Service t

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Legend:

Surface decarburization — — — — — — — — — —

Internal decarburization ————————

| | Carbon steel | 1.0Cr 0.5Mo | 2.25Cr 1.00Mo | 3.0Cr 1.0Mo | 6.0Cr 0.5Mo |
|---|---|---|---|---|---|
| Satisfactory | ○ | □ | ▭ | ◇ | ▽ |
| Internal decarburization and fissuring | ● | ■ | ▬ | ◆ | ▼ |
| Surface decarburization | ⊠ | ⊠ | ▨ | ⊠ | ⊠ |
| See comments | ◑ | ◧ | ◨ | ◈ | ▼ |

Avoid Decarburization and Fissuring

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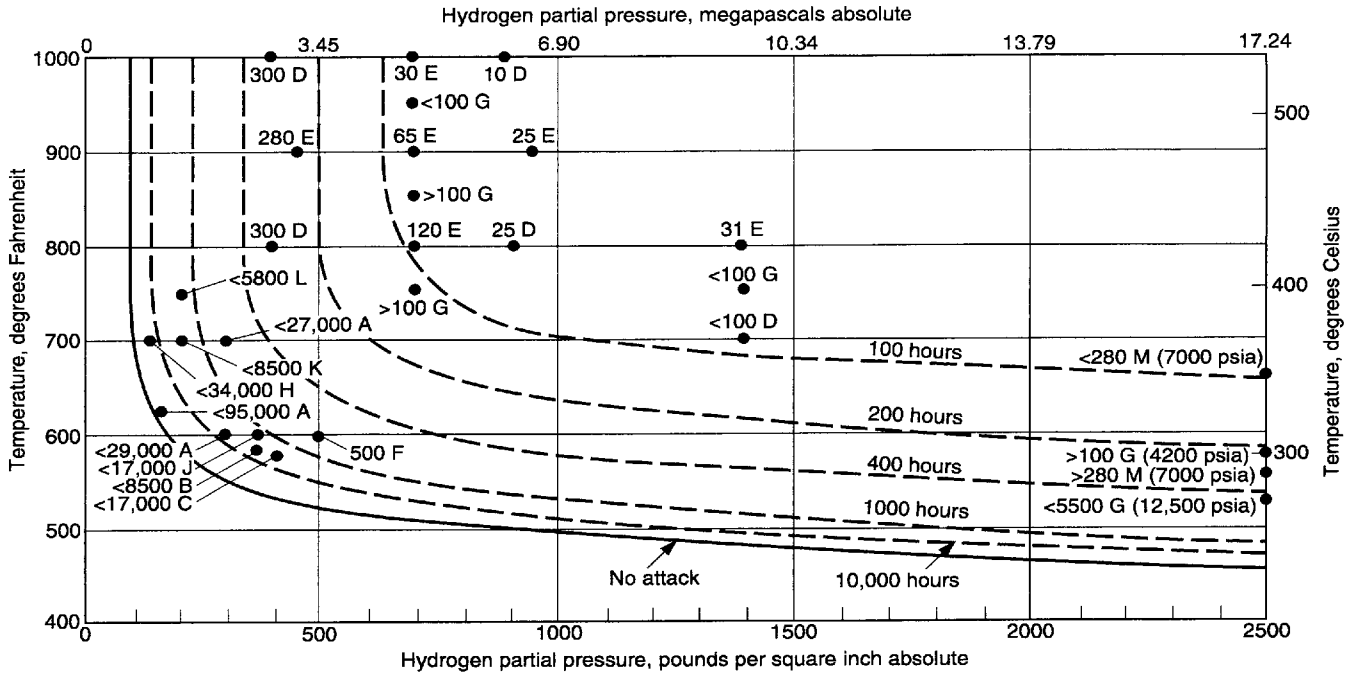


voids reach a critical size, the effects on mechanical properties become significant. The incubation period depends on many variables, including the type of steel, degree of cold working, amount of impurity elements, applied stress, hydrogen pressure, and temperature.

Incubation curves for carbon steel are given in Figure 3.[17] These can be used as a guide in determining approximate safe operating times for steels that operate above their long-term experience curves. Appendix A includes similar curves that may be useful for some heats of C–0.5Mo steel, with the precaution that resistance of C–0.5Mo steel to high temperature hydrogen attack is particularly sensitive to heat treatment, chemical composition, and the heating/cooling history of the steel during forming.[18, 19, 20, 21]

## 5.3 EFFECT OF PRIMARY STRESSES

Many users have reported satisfactory performance of annealed or normalized and tempered steels produced before 1969, as shown in Figure 1. These steels have been used for pressure-retaining equipment at design stress levels allowed by the 1969 or earlier editions of commonly-accepted codes (such codes include the ASME Code, Section VIII, Division 1; the standards of the American National Standards Institute; and, for the lower-strength materials, those of Deutsche Industrie-Normen). However, pressure vessels in hydrogen service have also been constructed using the higher allowable stresses permitted in either Section VIII, Division 2, or modifications of Section III of the ASME Code. Quenched and tempered or normalized and tempered steels have normally






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Figure 3—Time For Incipient Attack of Carbon Steel in High Temperature Hydrogen Service

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been utilized for these vessels due to their improved mechanical properties (strength and impact toughness).

No incidents of decarburization or fissuring of pressure vessels built to the specifications of Section VIII, Division 2, of the ASME Code have been reported. None of the failure data in Figure 1 represent materials used at the higher allowable stresses.

Recently published laboratory creep studies[22] have shown that the rupture strength and rupture ductility of 2.25Cr–1Mo steel are diminished when tested in hydrogen as compared to their values in air. These tests were a continuation of previously-reported tests[23, 24, 25] that showed somewhat conflicting results in shorter term tests.

These recent tests were conducted at applied stress levels similar to those that might be experienced by ASME Section VIII, Division 2 vessels. Test exposure times exceeded 50,000 hours depending on applied stress and temperature. The test specimens were from weldments of thick section plates and represented base metal, weld metal, and heat-affected zone. Detrimental effects of hydrogen were found down to the Figure 1 limit of 850°F (454°C) at 2000 pounds per square inch absolute (14 megapascals) and 3000 pounds per square inch absolute (21 megapascals) hydrogen partial pressure.

## 5.4 EFFECT OF SECONDARY STRESSES

High temperature hydrogen attack can be accelerated by secondary stresses, such as thermal stresses or those induced by cold work. High thermal stresses were considered to play a significant role in the high temperature hydrogen attack of some 2.25Cr–1Mo steel piping.[26] Other 2.25Cr–1Mo steel piping in the same system, subjected to more severe hydrogen partial pressures and temperatures, was not attacked.

The effect of cold work was demonstrated by Vitovec in work sponsored by API and summarized in API Publication 940.[9] Vitovec compared specific gravities of SAE 1020 steel with varying degrees of cold work tested in 900 psi (6.2 megapascals) hydrogen at 700°F (371°C), 800°F (427°C), and 1000°F (538°C). The decrease in specific gravity over time indicates the rate at which internal fissures produced by high temperature hydrogen attack are developed.

Annealed samples (0 percent strain) had an incubation period followed by a decrease in specific gravity. Steels with 5 percent strain had shorter incubation periods, and specific gravity decreased at a more rapid rate. Steels with 39 percent strain showed no incubation period at any test temperature, indicating that fissuring and cracking started immediately upon exposure to hydrogen.

These tests are considered significant in explaining the cracks sometimes found in highly stressed areas of an otherwise apparently resistant material. In addition, Cherrington and Ciuffreda[27] have emphasized the need for removing notches (stress concentrators) in hydrogen service equipment.

## 5.5 EFFECT OF HEAT TREATMENT

Both industry experience and research work indicate that postweld heat treatment (PWHT) of chromium-molybdenum steels in hydrogen service improves resistance to high temperature hydrogen attack. The PWHT stabilizes alloy carbides. This reduces the amount of carbon available to combine with hydrogen, thus improving high temperature hydrogen attack resistance.

Research[7,16,20, 21, 28] has shown that certain metal carbides may be more resistant to decomposition in high temperature hydrogen environments. Creep tests in hydrogen demonstrated the beneficial effect of increased PWHT on the high temperature hydrogen attack resistance of 2.25Cr–1Mo steel.[22] In these tests, 2.25Cr–1Mo steels postweld heat treated for 16 hours at 1275°F (691°C) showed more resistance to high temperature hydrogen attack than the same steels postweld heat treated for 24 hours at 1165°F (630°C). Both high PWHT temperatures and longer times are beneficial. Similarly, high temperature hydrogen attack resistance of 1Cr–0.5Mo and 1.25Cr–0.5Mo steels is improved by raising the minimum PWHT temperature to 1250°F (677°C) from the 1100°F (593°C) minimum required by Section VIII of the ASME Code.

The user must balance the advantages of high PWHT temperatures with other factors, such as the effect upon strength and notch toughness. Note higher PWHT temperatures can affect the ability to meet ASME Code Class 2 strength requirements.

## 5.6 EFFECT OF STAINLESS STEEL CLADDING OR WELD OVERLAY

The solubility of hydrogen in austenitic stainless steel is about an order of magnitude greater than for ferritic steels.[29] The diffusion coefficient of hydrogen through austenitic stainless steel is roughly two orders of magnitude lower than for ferritic steels.[30, 31] A sound, metallurgically bonded austenitic stainless steel cladding or weld overlay can reduce the effective hydrogen partial pressure acting on the base metal. Ferritic or martensitic claddings or weld overlays have little or no benefit in reducing the hydrogen partial pressure acting on the base metal.

The amount of hydrogen partial pressure reduction depends upon the materials and the relative thickness of the cladding/weld overlay and the base metal—the thicker the stainless barrier is relative to the base metal the better.[32] Archakov and Grebeshkova[33] mathematically considered how stainless steel corrosion barrier layers increase resistance of carbon and low alloy steels to high temperature hydrogen attack.

There have been a few instances of high temperature hydrogen attack of base metal which was clad or overlayed with austenitic stainless steel. All of the reported instances involved C–0.5Mo steel base metal. In one case,[34] high tem-

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# APPENDIX A—HIGH TEMPERATURE HYDROGEN ATTACK OF 0.5Mo STEELS

The purpose of this appendix is to provide a brief summary of the information and experience regarding the use of 0.5Mo (C–0.5Mo, Mn–0.5Mo) steels in elevated temperature and pressure hydrogen service.

Most companies no longer specify C–0.5Mo steel for new or replacement equipment used for operation above the carbon steel curve in Figure 1 because of the uncertainties regarding its performance after prolonged use. Since 1970, a series of unfavorable service experiences with C–0.5Mo steels has reduced confidence in the position of the 0.5Mo curve.[A1, A2] In the second edition (1977) of this publication, the 0.5Mo curve was lowered approximately 60°F (33°C) to reflect a number of plant experiences that involved high temperature hydrogen attack of C–0.5Mo equipment. In the fourth edition (1990) of this publication, the 0.5Mo curve was removed from Figure 1 due to additional cases of high temperature hydrogen attack of C–0.5Mo steel equipment as much as 200°F (111°C) below the curve. Plant experience has identified 27 instances of high temperature hydrogen attack below the 1977 curve. The operating conditions for these instances are given in Table A-1, and are plotted on Figure A-1.

No instances have been reported of high temperature hydrogen attack of Mn–0.5Mo steel below the Figure A-1 0.5Mo curve. The information and use of this material at elevated temperatures and hydrogen partial pressures are limited.

C–0.5Mo steels vary in their resistance to high temperature hydrogen attack. Many heats seem to have resistance at conditions indicated by the 0.5Mo curve on Figure A-1. However, some heats seem to have high temperature hydrogen attack resistance only marginally better than carbon steel. Recent work[A2,A3,A4,A5] relating its resistance to high temperature hydrogen attack to the thermal history of the steel give some guidance as to why. Slow-cooled, annealed C–0.5Mo steels have less resistance to high temperature hydrogen attack than normalized steels. The studies have shown that postweld heat treatment improves the high temperature hydrogen attack resistance of weldments and heat affected zones for both annealed and normalized C–0.5Mo steels. However, the base metals of slow-cooled, annealed C–0.5Mo steels show a decrease in high temperature hydrogen attack resistance after postweld heat treatment. The initial studies suggest that this is due to free carbon being present in the ferrite matrix after postweld heat treatment. Normalized C–0.5Mo steel base metals, on the other hand, show improvement in high temperature hydrogen attack resistance following tempering or postweld heat treatment. Such normalized and postweld heat treated C–0.5Mo steel appears to have hydrogen attack resistance about as indicated by the 0.5Mo curve in the second edition (1977) of this publication. Until the factors controlling the high temperature hydrogen attack resistance of C–0.5Mo are better understood, each user should carefully assess the use of C–0.5Mo steel in services above the carbon steel curve in Figure A-1.

Existing C–0.5Mo steel equipment that is operated above the carbon steel curve in Figure A-1 should be inspected to detect high temperature hydrogen attack. Owners/operators should evaluate and prioritize C–0.5Mo equipment operating above the carbon steel limit for inspection—Hattori and Aikawa[A6] addressed this issue. The work cited above and plant experience suggest that important variables to consider in prioritizing equipment for inspection include severity of operating condition (hydrogen partial pressure and temperature), thermal history of the steel during fabrication, stress, cold work, and cladding composition and thickness, when present.

To provide a historical summary of the data regarding the use of C–0.5Mo steels, two additional figures are included here: (1) Figure A-2 which shows the effect of trace alloying elements (molybdenum) on operating limits, and (2) Figure A-3 which shows incubation times for C–0.5Mo steels. Figure A-2 is from the second edition of this publication (1977), and is a revision of a similar figure from the original edition (1970). Figure A-2 shows that molybdenum has long been considered to be beneficial to the high temperature hydrogen attack resistance of steels. The data in Figures A-2 and A-3 should be used with caution, since some heats of C–0.5Mo steels have suffered high temperature hydrogen attack during exposure to conditions under the lower solid curve (equivalent to the C–0.5Mo curve of Figure A-1). The data for the instances of high temperature hydrogen attack listed in Table A-1 and plotted on Figure A-1 are also shown for reference in Figure A-3. The incubation time would be less than the service life at the time the attack was detected.

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## REFERENCES AND COMMENTS FOR FIGURE A-1

The data in Figure A-1 are labeled with reference numbers corresponding to the sources listed below. The letters in the figure correspond to the comments listed on this page.

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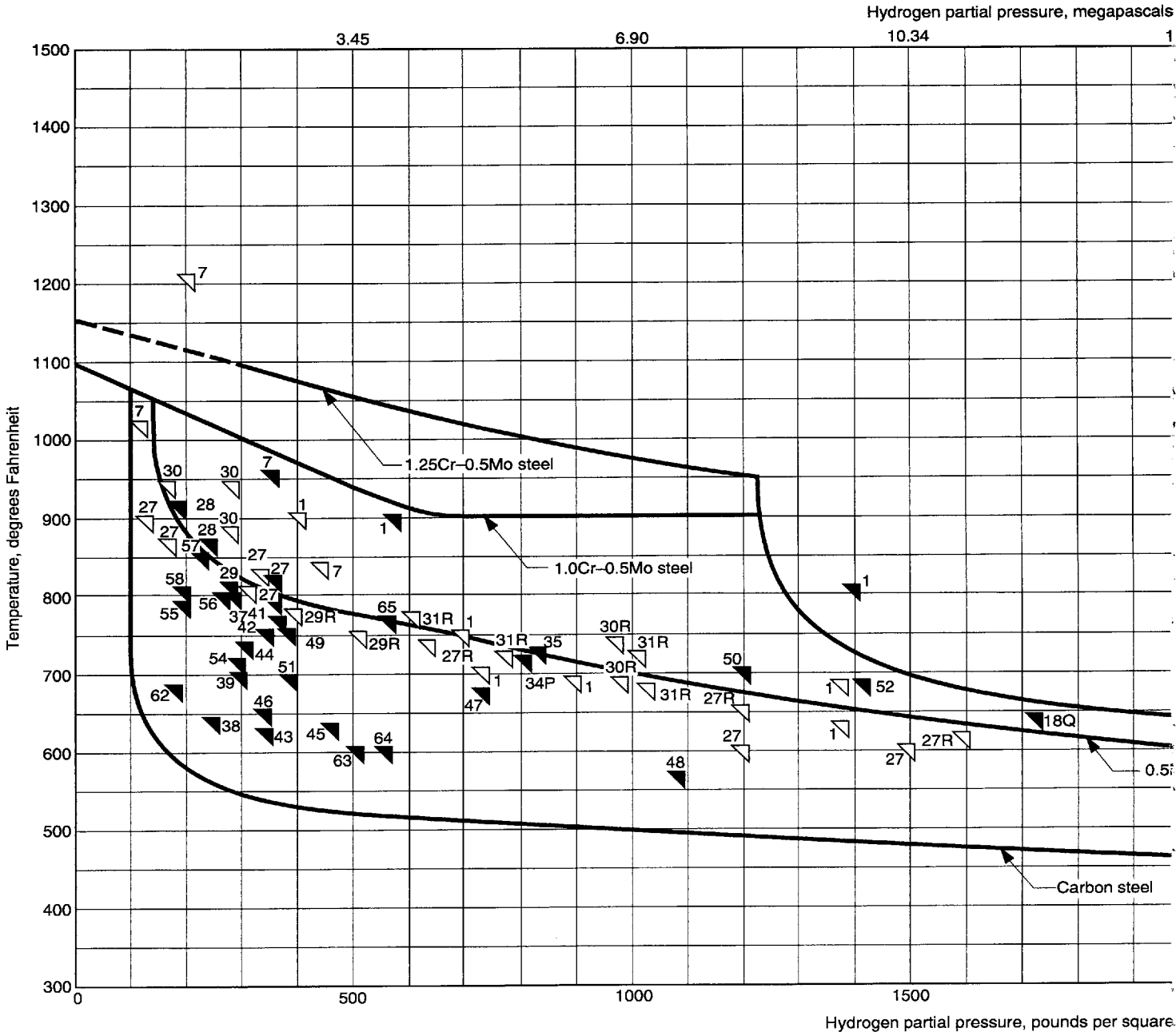

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65. [LL]Exxon report: "Hydrogen Attack of Gofiner Reactor Inlet Nozzle," 1988.


### Comments

A. Feed line pipe leaked; isolated areas damaged. Blistered, decarburized, fissured; postweld heat treated at 1100°F to 1350°F.
B. Effluent line, pipe and heat-affected zone, isolated areas damaged; no postweld heat treatment.
C. Weld and pipe, isolated areas damaged; no postweld heat treatment.
D. Effluent line; weld, isolated areas damaged; postweld heat treatment.
E. Feed line; weld and heat-affected zone, isolated areas damaged; postweld heat treatment.
F. Feed/effluent exchanger nozzle-to-shell weld, cracks in welds and in exchanger tubes.
G. Effluent exchanger channel; welds, plate, and heat-affected zone, isolated areas damaged; postweld heat treatment.
H. Effluent exchanger channel; welds, plate, and heat-affected zone, isolated areas damaged; postweld heat treated at 1100°F.
I. Catalytic reformer, combined feed/effluent exchanger shell; plate; postweld heat treated at 1250°F.
J. Hydrodesulfurization unit effluent exchanger channel head and shell plate. (Hydrocarbon feed to unit and make-up hydrogen from ethylene unit.)
K. Catalytic reformer combined feed piping; welds and base metal; postweld heat treatment.
L. Gas-oil hydrodesulfurization unit. Elbow cracked intergranularly and decarburized at fusion line between weld metal and heat-affected zone; no postweld heat treatment.
M. Ammonia plant converter; exit piping; intergranular cracking and internal decarburization of pipe.
P. Hydrodesulfurization unit hydrogen preheat exchanger shell; blisters, intergranular fissuring, and decarburization in weld metal; postweld heat treated at 1150°F.
Q. Attack of heat exchanger tubing in tubesheet.
R. Stainless steel cladding on 0.5Mo steel; no known HTHA.
S. Decarburization and fissuring of weld metal; postweld heat treated at 1150°F.
T. Forged tubesheet cracked with surface decarburation; tubes blistered.
U. Hydrodesulfurization unit, C-0.5Mo steel exchanger tubesheet; decarburized, fissured, and cracked under intergranularly cracked ASTM Type 304 cladding.
V. Hydrocracker charge exchanger liquid with a small amount of hydrogen; C-0.5Mo with Type 410S rolled bond clad. Extensive blistering and fissuring under clad.
W. C-0.5Mo steel piping in ammonia plant syngas loop; decarburized and fissured.
AA. Blistering and fissuring of a flange.
BB. HAZ and base metal fissuring of pipe.
CC. Base metal fissuring and surface blistering in heat exchanger shell.
DD. Attack at weld, HAZ and base material in piping.
EE. Localized attack in weld, HAZ in piping.
FF. Base metal attack in piping.
GG. Base metal attack in a heat exchanger channel.
HH. Base metal attack in piping.
II. Blistering and base metal attack in a heat exchanger shell.
JJ. Base metal attack in a TP405 roll bond clad vessel.
KK. Base metal attack in a TP405 roll bond clad vessel.
LL. Attack in nozzle attachment area of a vessel weld overlaid with Type 309Nb.
MM. Internal decarburization/fissuring of piping in a hydrocracker unit after 235,000 hours of service.



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Notes:
1. References and comments are shown on Table A-1.
2. Curves for carbon steel, 1.0Cr–0.5Mo steel, and 1.25Cr–0.5Mo steel are included for reference.
3. The symbol ◹ is retained as a reference against previous revisions of this publication.
4. Reference numbers are the same as in previous editions of this publication.
5. The 0.5Mo steel curve is the same as the one shown in the fourth edition of this publication (1990).

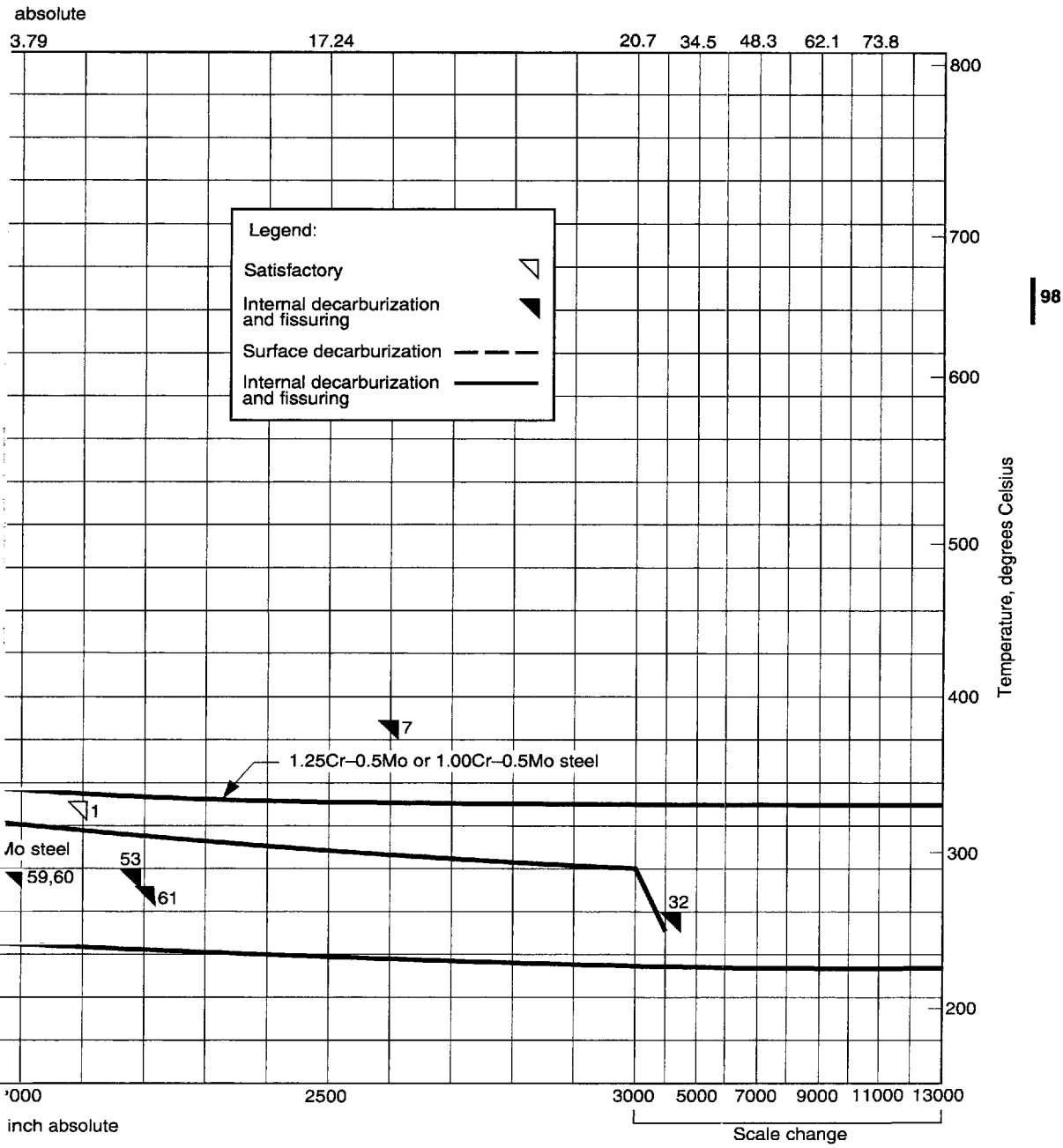
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This figure was revised by API in 1969, 1977, 1983, 1990 and 1996.

API 941 Fifth Edition
January 1997

Figure A-1—Experience with C–0.5Mo and Mn–0.5Mo Steel in



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High Temperature Hydrogen Service

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## Table A-1—Operating Conditions For C–0.5Mo Steels That Experienced High Temperature Hydrogen Attack Below the 0.5Mo Steel Curve in Figure A-1

| Point | Temperature | | Hydrogen Partial Pressure (Absolute) | | Service Hours (Approx.) | Degrees Below 0.5Mo Curve (Approximate) | | |
|---|---|---|---|---|---|---|---|---|
| | °F | °C | psi | MPa | | °F | °C | |
| 36A[1] | 790 | 421 | 350 | 2.41 | 80,000 | 20 | 11 | |
| 37B[1] | 800 | 427 | 285 | 1.97 | 57,000 | 30 | 17 | |
| 38C[1] | 640 | 338 | 270 | 1.86 | 83,000 | 180 | 100 | |
| 39D[1] | 700 | 371 | 300 | 2.07 | 96,000 | 125 | 69 | |
| 41F[1] | 760 | 404 | 375 | 2.59 | 85,000 | 40 | 22 | |
| 42G[1] | 750 | 399 | 350 | 2.41 | 150,000 | 60 | 33 | |
| 43H[1] | 625 | 329 | 350 | 2.41 | 150,000 | 185 | 103 | |
| 44I[1] | 730[2] | 388[2] | 313 | 2.16 | 116,000 | 90 | 50 | |
| 45J[3] | 620/640 | 327/338 | 457 | 3.15 | 70,000 | 167/147 | 93/82 | |
| 46K[1] | 626/680 | 330/360 | 350 | 2.41 | 131,000 | 184/130 | 102/72 | |
| 47L[3] | 684[2] | 362 | 738 | 5.09 | 61,000 | 54 | 30 | |
| 48M[5] | 550/570 | 288/299 | 1060/1100 | 7.31/7.59 | 79,000 | 125/105 | 69/58 | |
| * | 655/670 | 346/354 | - | - | 17,500 | 20/5 | 11/3 | |
| 49S[3] | 750/770 | 399/410 | 390 | 2.69 | 67,000 | 50/30 | 28/17 | |
| * | 650 | 343 | - | - | 163,000 | 150 | 83 | |
| 51U[3] | 690 | 366 | 397 | 2.74 | - | 100 | 56 | |
| 53W[5] | 545 | 285 | 2190 | 15.1 | 140,000 | 45 | 25 | |
| 54AA[1] | 725/760 | 385/404 | 300/380 | 2.07/2.62 | 105,000 | 40/100 | 22/56 | 98 |
| 55BB[1] | 800/850 | 427/454 | 175/190 | 1.21/1.31 | 124,000 | 80/30 | 44/17 | |
| 56CC[1] | 810/825 | 432/441 | 275/300 | 1.90/2.07 | 223,000 | 15/0 | 8/0 | |
| 57DD[1] | 850[4] | 454[4] | 225[4] | 1.55[4] | 158,000 | 10 | 6 | |
| 58EE[1] | 810/855 | 432/457 | 170 | 1.17 | 138,000 | 70/25 | 39/14 | |
| 59FF[3] | 550/600 | 288/316 | 2000 | 13.79 | 210,000 | 50/0 | 28/0 | |
| 60GG[3] | 550/600 | 288/316 | 2000 | 13.79 | 210,000 | 50/0 | 28/0 | |
| 61HH[3] | 530/600 | 277/316 | 2200 | 15.17 | 210,000 | 60/0 | 33/0 | |
| 62II[3] | 670/700 | 354/371 | 190 | 1.31 | 192,000 | 180/150 | 100/83 | |
| 63JJ[3] | 600/750 | 316/399 | 500 | 3.45 | 235,000 | 180/30 | 100/17 | |
| 64KK[3] | 600/770 | 316/410 | 525 | 3.62 | 283,000 | 170/0 | 94/0 | |
| 65LL[3] | 775 | 413 | 550 | 3.79 | - | 0 | 0 | |

Notes:

Numbers and letters in the first column (labeled "Point") refer to references and comments for Figure A-1.

Where two numbers are given, the first number represents average operating conditions while the second number represents maximum operating conditions. 98

[1]Catalytic reformer service.

[2]Average.

[3]Hydrodesulfurizer service.

[4]Maximum.

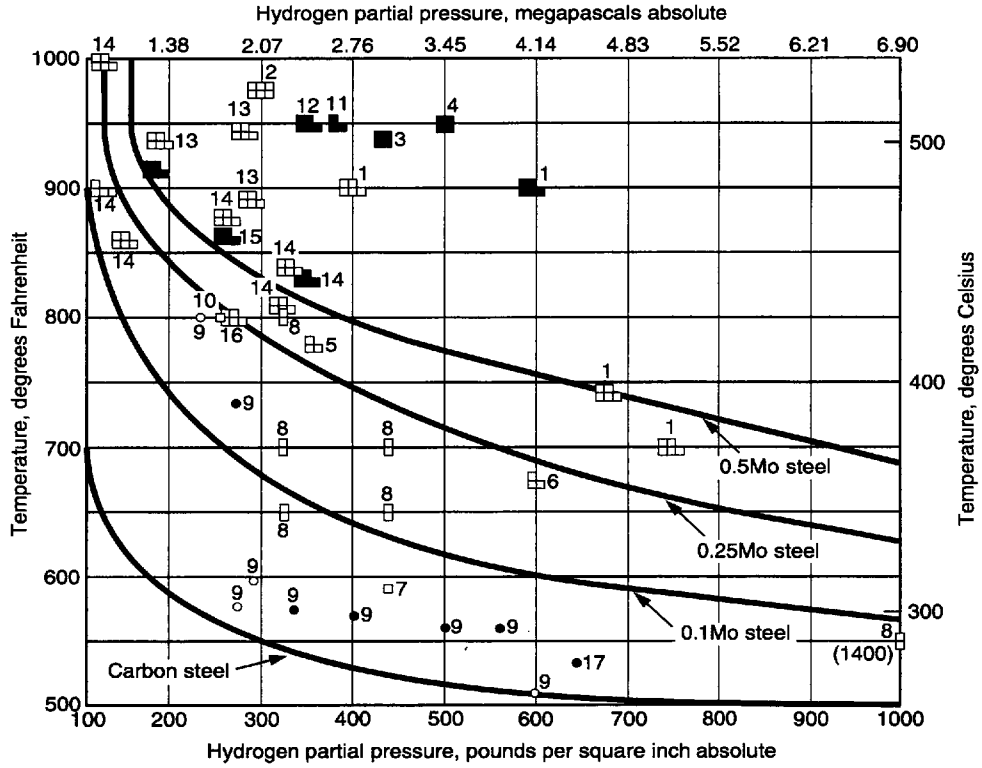
[5]Ammonia plant.

* API TG currently resolving these points.

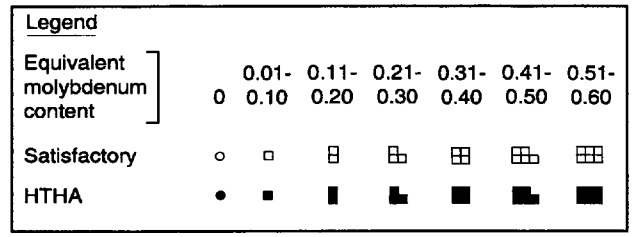
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Notes
1. Mo has four times the resistance of Cr to HTHA.
2. Mo is equivalent to V, Ti, or Nb up to 0.1 percent
3. Si, Ni, and Cu do not increase resistance.
4. P, and S decrease resistance.

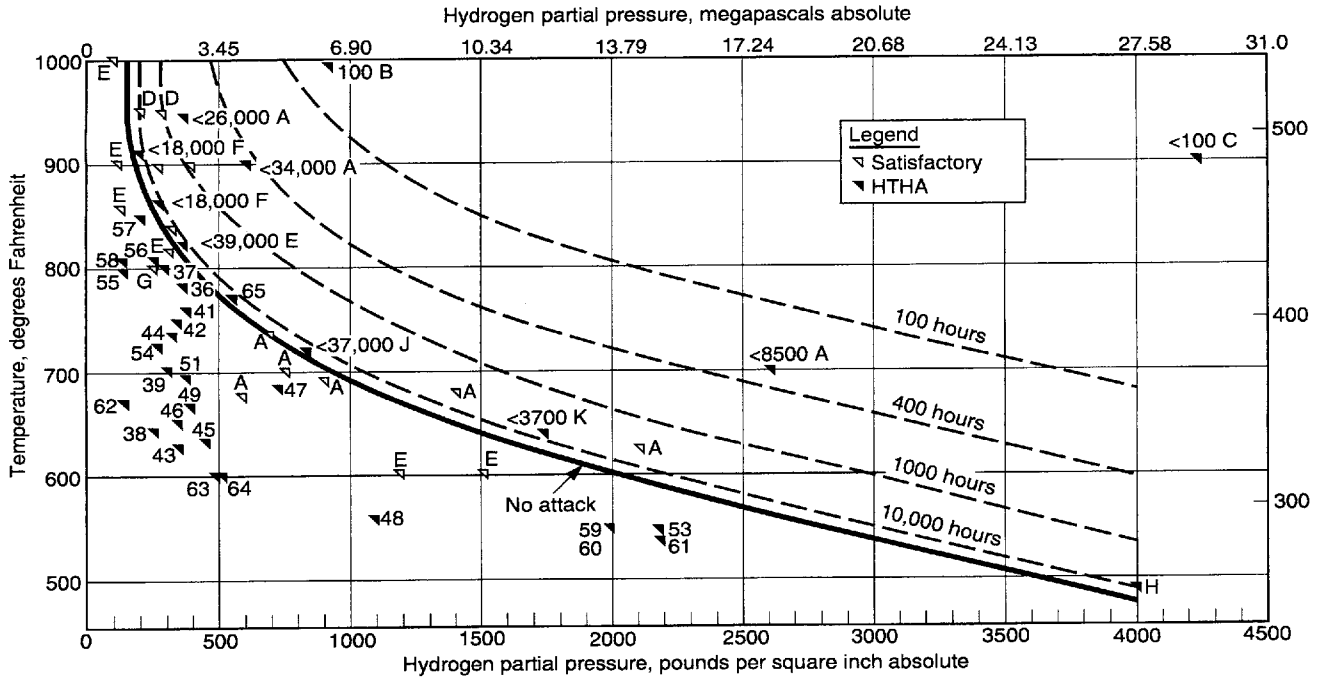



| References | | Analysis Cr | Mo | V | Mo Equiv. |
|---|---|---|---|---|---|
| 1. | Shell Oil Company.* | | 0.50 | | 0.50 |
| 2. | Weld Deposits, D. J. Bergman.* | 0.79 | 0.39 | | 0.59 |
| 3. | Weld Deposits, D. J. Bergman.* | 0.80 | 0.15 | | 0.35 |
| 4. | Weld Deposits, D. J. Bergman.* | 0.50 | 0.25 | | 0.37 |
| 5. | Continental Oil Company.* | | 0.25 | | 0.25 |
| 6. | Standard Oil Co. of California.* | | 0.27 | | 0.27 |
| 7. | Standard Oil Co. of California.* | 0.05 | 0.06 | 0.08 | |
| 8. | A. O. Smith Corp.* | | | 0.13-0.18 | 0.11 |
| 9. | Shell Development Co., Drawing No. VT 659-2. | | | | |
| 10. | Amoco Oil Company.* | 0.04 | | | 0.01 |
| 11. | R. W. Manuel, Corrosion, 17(9), pp. 103-104, Sept. 1961 | 0.27 | 0.15 | | 0.22 |
| 12. | The Standard Oil Co. of Ohio.* | 0.11 | 0.43 | | 0.50 |
| 13. | Exxon Corporation.* | | | | |
| 14. | Union Oil of California.* | | | | |
| 15. | Amoco Oil Company.* | | | | |
| 16. | Standard Oil Co. of California. * | | | | |
| 17. | Gulf Oil Corporation.* | | | | |

*Private communication to Subcommittee on Corrosion (now Subcommittee on Corrosion and Materials).

Figure A-2—Steels in High Temperature Hydrogen Service Showing Effect of Molybdenum and Trace Alloying Elements

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Notes
1. This figure was adapted from Figure 3, Fourth Edition (1990) of this publication.
2. Numbered and lettered references for points in this figure refer to data listed in Table A-1 and Figure A-1 comments.

Figure A-3—Time For Incipient Attack of 0.5Mo Steels in High Temperature Hydrogen Service

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# APPENDIX B—HIGH TEMPERATURE HYDROGEN ATTACK OF 1.25 Cr-0.5Mo STEEL

The purpose of this appendix is to provide a brief summary of the information and experience regarding three case histories with high temperature hydrogen attack of 1.25Cr–0.5Mo steel.

The three recent experiences with high temperature hydrogen attack are listed in Table B-1, and the operating conditions are plotted in Figure B-1.

Table B-1—Experience with High Temperature Hydrogen Attack of 1.25Cr–0.5Mo Steel at Operating Conditions Below the Figure 1 Curve

| Case | Temperature °F | Temperature °C | Hydrogen Partial Pressure (Absolute) psi | Hydrogen Partial Pressure (Absolute) MPa | Service Years | Description |
|---|---|---|---|---|---|---|
| A | 960 | 516 | 331 | 2.28 | 26 | 1.5 NPS Schedule 80 nozzle was broken off a catalytic reformer outlet line during a shut down. Metallography indicated surface decarburization and intergranular cracking with bubbles. Cr content was 1.09 percent. |
| B | 977 | 525 | 354 | 2.44 | 14 | Blistering was detected with ultrasonic examination in catalytic reformer piping. Metallography indicated surface decarburization and blistering at non-metallic inclusions, with intergranular cracks growing from the blisters. Cr content was 1.10 percent. |
| C | 957/ 982 | 514/ 528 | 294/ 408 | 2.03/ 2.81 | 16 | Blistering near pipe inner surface. Examination showed decarburization between the inner surface and the blister. Gas analysis indicated methane in the blister. Cr content was 1.12 percent. |
| | Note 1 | Note 1 | Note 1 | Note 1 | | |

Note 1: Average conditions are reported as the left number. Maximum condition reported as the right number.

98

Cases A and B were reported by Chiyoda Corporation in Japan. Case C was originally reported by Merrick and Maguire of Exxon (see reference 10 in 2.2). The mechanisms of attack were similar in Cases B and C. That is, damage was in the form of internal blistering, with decarburization and intergranular cracking from the edges of the blisters. In Case A, however, attack resulted in intergranular separation. All three steels had chromium contents near 1.1 percent, near the 1.0 percent lower limit for 1.25Cr–0.5Mo steels. Additionally, the Case A steel had a relatively high impurity content with an $\bar{x}$ equal to 31.5, where $\bar{x}$ is defined as:

$$\bar{x} = \frac{10P + 5Sb + 4Sn + As}{100}, \text{ all elements in ppm.}$$ [37]



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98

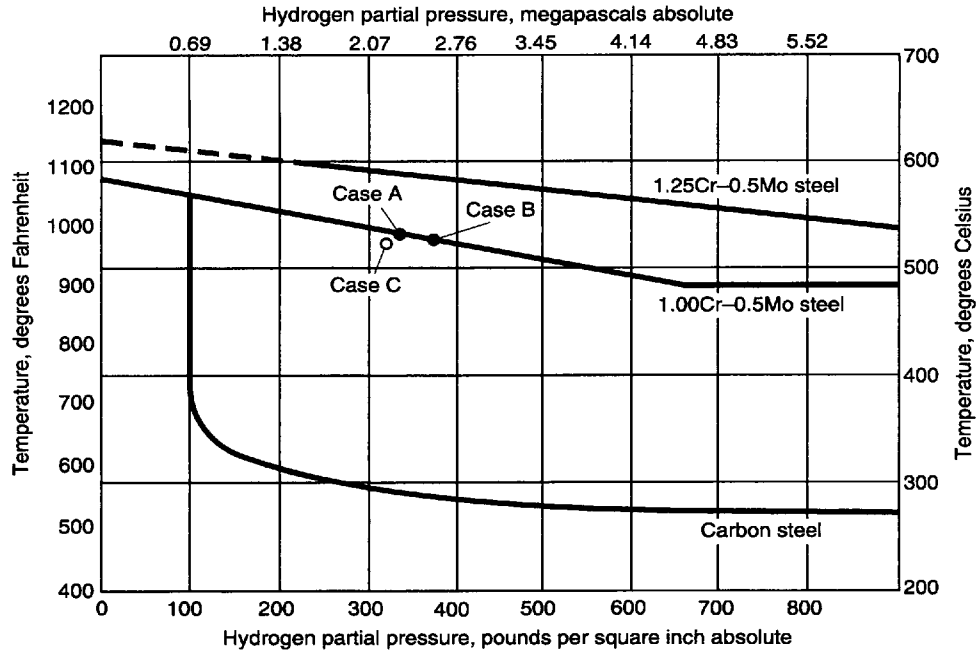



Figure B-1—Operating Conditions for 1.25Cr–0.5Mo Steels That Experienced HTHA Below the Figure 1 Curve

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# APPENDIX C—HIGH TEMPERATURE HYDROGEN ATTACK OF 2.25Cr-1Mo STEEL

The purpose of this appendix is to provide a brief summary of experience regarding a case history[a] with high temperature hydrogen attack of 2.25Cr–1Mo steel.

A recent experience with high temperature hydrogen attack is described in Table C-1, and the operating conditions are plotted in Figure C-1.

This case history may indicate that highly stressed components can suffer high temperature hydrogen attack at conditions below the curve in Figure 1. In this case history, the mixing tee was believed to be highly stressed by thermal stresses due to the mixing of hot and cooler hydrogen. Figure C-1 shows the operating conditions of both the hot upstream hydrogen and the mixed hydrogen downstream of the tee.

Table C-1—Experience With High Temperature Hydrogen Attack of 2.25Cr–1Mo Steel at Operating Conditions Below the Figure 1 Curve

| Temperature | | Hydrogen Partial Pressure (Absolute) | | Time in Service Years | Description |
|---|---|---|---|---|---|
| °F | °C | psi | MPa | | |
| 675/820 | 357/438 | 1385/1570 | 9.54/10.82 | >20 | A mixing tee for the hot and cold make-up hydrogen to a hydroprocessing unit leaked near the weld to the downstream piping. SEM examination indicated decarburization and fissuring along the internal surface of the tee. |
| See Note | See Note | See Note | See Note | | Although the leak path was not positively identified, it was concluded to be most likely due to fine, interconnected fissures. Some thermal fatigue cracking was also identified in the tee. Piping downstream of the tee was also found to have fissuring and internal decarburization to a depth of about 3.90 mils (0.1 mm) along the inside surface. The hot, upstream piping was not found to be attacked. |

Note: Average conditions are reported as first number. Maximum condition reported as second number.

[a]Communication to the API Subcommittee on Corrosion and Materials from Exxon Research and Engineering, August 1995.



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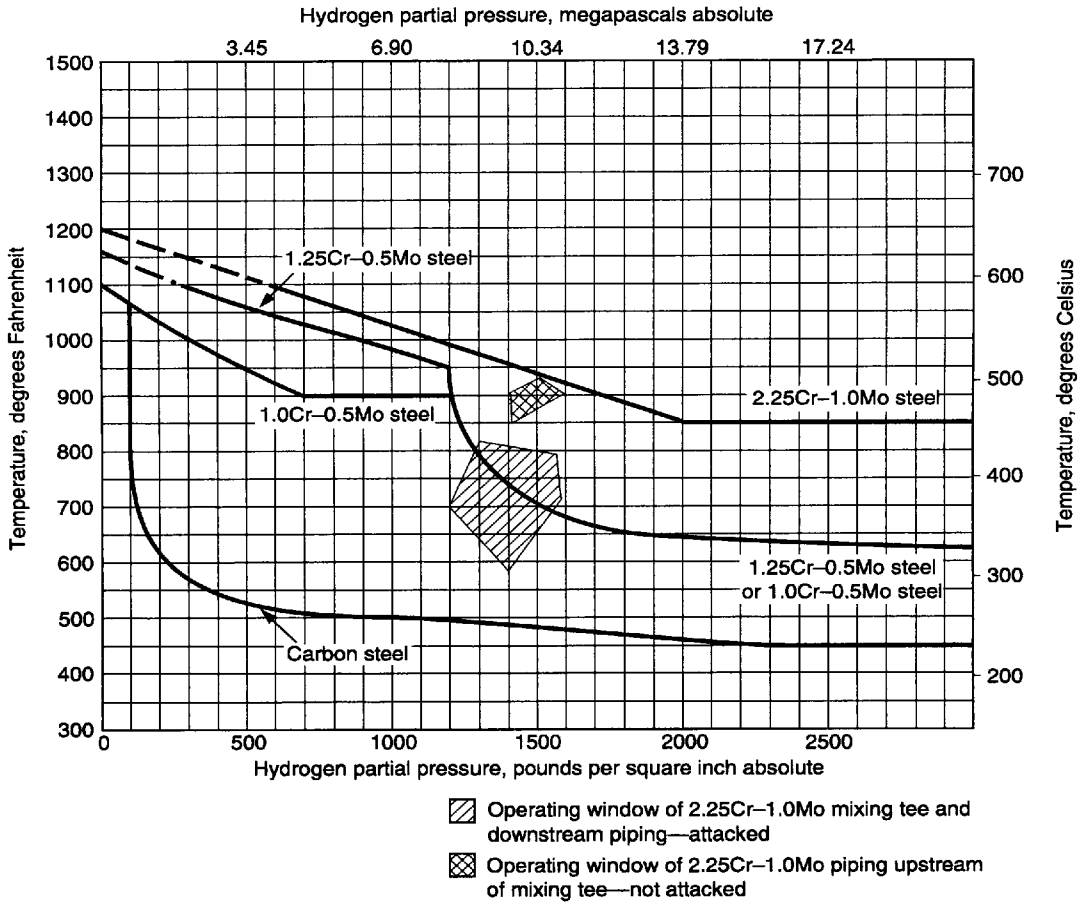



Figure C-1—Operating Conditions of 2.25Cr–1Mo Steels That Experienced HTHA Below the Figure 1 Curve

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**Affected Publication:** API Recommended Practice 941, *Steels For Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants*, 5th Edition, January 1997

## ERRATA

*Pages 5 & 15, Figures 1 and A-1:*
Electronic transcription of these figures resulted in slight shifts in the trace of the curves when compared to the fourth edition of RP 941. There was no intention to change the curve positions in the fifth edition of RP 941.

*Page 14, References and Comments for Figure A-1:*
Under References, change date on reference number 57 from "June 1947" to "June, 1992".

*Page 15, Figure A-1:*
Change point 37 (at 850°F) to "57". The triangle at point 56 should be blackened. In the legend, change "Hydrogen attack" to: "Internal decarburization and fissuring".

*Page 17, Table A-1, add general note:*
"Where two numbers are given, the first number represents average operating conditions while the second number represents maximum operating conditions."

*Page 19, Figure A-3, in the legend, change:*
"No attack" to: Satisfactory

*Page 21, Table B-1, Note 1:*
Replace the word "top" in the first sentence with "left", and replace the word "bottom" in the second sentence with "right".

*Page 24, Figure C-1, the numbers on the bottom axis for hydrogen partial pressure should read:*
"0", "500", "1000", "1500", "2000", and "2500".

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# Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants

API RECOMMENDED PRACTICE 941
FIFTH EDITION, JANUARY 1997



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# Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants

**Manufacturing, Distribution and Marketing Department**

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# CONTENTS


| | | Page |
|---|---|---|
| 0 | INTRODUCTION | 1 |
| 1 | SCOPE | 1 |
| 2 | REFERENCES | 1 |
| | 2.1 Standards | 1 |
| | 2.2 Other References | 2 |
| 3 | OPERATING LIMITS | 3 |
| | 3.1 Bases For Setting Operating Limits | 3 |
| | 3.2 Selecting Materials For New Equipment | 3 |
| 4 | FORMS OF HIGH TEMPERATURE HYDROGEN ATTACK | 7 |
| | 4.1 General | 7 |
| | 4.2 Surface Decarburization | 7 |
| | 4.3 Internal Decarburization and Fissuring | 7 |
| 5 | FACTORS INFLUENCING HTHA | 8 |
| | 5.1 High Temperature Hydrogen Attack In A Liquid Hydrocarbon Phase | 8 |
| | 5.2 Incubation | 8 |
| | 5.3 Effect of Primary Stresses | 9 |
| | 5.4 Effect of Secondary Stresses | 10 |
| | 5.5 Effect of Heat Treatment | 10 |
| | 5.6 Effect of Stainless Steel Cladding or Weld Overlay | 10 |
| 6 | INSPECTION | 11 |
| | 6.1 Inspection For High Temperature Hydrogen Attack | 11 |
| APPENDIX A—HIGH TEMPERATURE HYDROGEN ATTACK OF 0.5Mo STEELS | | 13 |
| APPENDIX B—HIGH TEMPERATURE HYDROGEN ATTACK OF 1.25Cr–0.5Mo STEEL | | 21 |
| APPENDIX C—HIGH TEMPERATURE HYDROGEN ATTACK OF 2.25Cr–1Mo STEEL | | 23 |
| APPENDIX D—INSPECTION FOR HTHA TABLES | | 25 |
| APPENDIX E—REQUEST FOR NEW INFORMATION | | 31 |

Figures

1—Operating Limits for Steels in Hydrogen Service to Avoid Decarburization and Fissuring ..... 5
2—C–0.5Mo Steel Showing Internal Decarburization and Fissuring in High Temperature Hydrogen Service ..... 8
3—Time for Incipient Attack of Carbon Steel in High Temperature Hydrogen Service ..... 9
A-1—Experience with C–0.5Mo and Mn–0.5Mo Steel ..... 15
A-2—Steels in High Temperature Hydrogen Service Showing Effect of Molybdenum and Trace Alloying Elements ..... 18
A-3—Time for Incipient Attack of 0.5Mo Steels in High Temperature Hydrogen Service ..... 19
B-1—Operating Conditions for 1.25Cr–0.5Mo Steels That Experienced HTHA Below the Figure 1 Curve ..... 22




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Page

C-1—Operating Conditions of 2.25Cr–1Mo Steels That Experienced HTHA Below the Figure 1 Curve ..... 24

Tables

A-1—Operating Conditions for C–0.5Mo Steels That Experienced High Temperature Hydrogen Attack Below the 0.5Mo Steel Curve in Figure A-1 ..... 17

B-1—Experience with High Temperature Hydrogen Attack of 1.25Cr–0.5Mo Steel at Operating Conditions Below the Figure 1 Curve ..... 21

C-1—Experience with High Temperature Hydrogen Attack of 2.25Cr–1Mo Steel at Operating Conditions Below the Figure 1 Curve ..... 23

D-1—Summary of Inspection Methods for High Temperature Hydrogen Attack ..... 27

D-2—Summary of Inspection Methods for High Temperature Hydrogen Attack ..... 29



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# Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants

## 0 Introduction

This recommended practice discusses the resistance of steels to high temperature hydrogen attack (HTHA) . In addition to this high temperature hydrogen attack, steels can be damaged at low temperatures from hydrogen entering the steel from aqueous corrosion mechanisms or from metal-working operations such as welding or electroplating. The following forms of low temperature hydrogen damage are defined and discussed in other industry publications. [1,2,3]

a. Hydrogen Embrittlement (HE).
b. Hydrogen Stress Cracking (HSC).
c. Sulfide Stress Cracking (SSC).
d. Hydrogen Blistering.
e. Hydrogen-Induced Cracking (HIC).
f. Stress-Oriented Hydrogen-Induced Cracking (SOHIC).

At normal atmospheric temperatures, gaseous molecular hydrogen does not readily permeate steel, even at high pressures. Carbon steel is the standard material for cylinders that are used to transport hydrogen at pressures of 2000 psi (14 megapascals). Many postweld heat treated carbon steel pressure vessels have been used successfully in continuous service at pressures up to 10,000 psi (69 megapascals) and temperatures up to 430°F (221°C). However, under these same conditions, highly-stressed carbon steel and hardened steels have cracked due to hydrogen embrittlement while operating at high hydrogen pressures and moderate temperatures.

The recommended maximum hydrogen partial pressure at atmospheric temperature for carbon steel fabricated in accordance with the ASME *Boiler and Pressure Vessel Code* is 13,000 pounds per square inch absolute (90 megapascals). Below this pressure, carbon steel equipment has shown satisfactory performance as long as the low temperature hydrogen damage mechanisms noted above are not involved. Above this pressure, very little operating and experimental data are available. If plants are to operate at hydrogen partial pressures that exceed 13,000 pounds per square inch absolute (90 megapascals), the use of an austenitic stainless steel liner with venting in the shell should be considered.

At elevated temperatures, molecular hydrogen dissociates into the atomic form, which can readily enter and diffuse through the steel. Under these conditions, the diffusion of hydrogen in steel is more rapid. As discussed in Section 4, Forms of High Temperature Hydrogen Attack, hydrogen may react with the carbon in the steel to cause either surface decarburization or internal decarburization and fissuring. This form of hydrogen damage is called high temperature hydrogen attack.

## 1 Scope

This recommended practice summarizes the results of experimental tests and actual data acquired from operating plants to establish practical operating limits for carbon and low alloy steels in hydrogen service at elevated temperatures and pressures. The effects on the resistance of steels to hydrogen at elevated temperature and pressure that result from high stress, heat treating, chemical composition, and cladding are discussed.

This recommended practice applies to equipment in refineries, petrochemical facilities, and chemical facilities in which hydrogen or hydrogen-containing fluids are processed at elevated temperature and pressure. The guidelines in this recommended practice can also be applied to hydrogenation plants such as those that manufacture ammonia, methanol, edible oils, and higher alcohols.

Hydrogenation processes usually require standards and materials that may not be warranted in other operations of the petroleum industry. At certain combinations of elevated temperature and hydrogen partial pressure, both chemical and metallurgical changes occur in carbon steel, which in advanced stages can render it unsuitable for safe operation. Alloy steels containing chromium and molybdenum can be used under such conditions.

The steels discussed in this recommended practice resist high temperature hydrogen attack (HTHA) when operated within the guidelines given. However, they may not be resistant to other corrosives present in a process stream or to other metallurgical damage mechanisms operating in the high temperature hydrogen attack range. This recommended practice also does not address the issues surrounding possible damage from rapid cooling of the metal after it has been in high temperature, high pressure hydrogen service (e.g., possible need for outgassing hydroprocessing reactors). This recommended practice will discuss in detail only the resistance of steels to high temperature hydrogen attack.

## 2 References

### 2.1 STANDARDS

Unless otherwise specified, the most recent editions or revisions of the following codes shall, to the extent specified herein, form a part of this publication.



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ASME[a]

*Boiler and Pressure Vessel Code, Section II,* "Materials" (Part A, "Ferrous Material Specifications," and Part D, "Properties"), Section III, "Rules for Construction of Nuclear Power Plant Components," and Section VIII, "Pressure Vessels," Divisions 1 and 2.

*Code for Pressure Piping,* ASME/ANSI B31.3, "Chemical Plant and Petroleum Refinery Piping"

## 2.2 OTHER REFERENCES

In addition, this publication draws upon the work referenced in the following publications:


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2. NACE International Technical Committee Report 8X194, "Materials and Fabrication Practices for New Pressure Vessels Used in Wet $H_2S$ Refinery Service."
3. NACE International Standard RP0472-95, *Methods and Controls to Prevent In-Service Environmental Cracking of Carbon Steel Weldments in Corrosive Petroleum Refining Environments.*
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# 3 Operating Limits

## 3.1 BASES FOR SETTING OPERATING LIMITS

Figure 1 illustrates the resistance of steels to attack by hydrogen at elevated temperatures and hydrogen pressures. As explained below, high temperature hydrogen attack of steel can result in surface decarburization, internal decarburization and fissuring, or both. Figure 1 gives the operating conditions (process temperature and hydrogen partial pressure) above which these types of damage can occur.

Figure 1 is based upon experience gathered since the 1940s. Supporting data were obtained from a variety of commercial processes and laboratory experiments (see the references to Figure 1). While temperature and hydrogen partial pressure data were not always known precisely, the accuracy is sufficient for commercial use. Satisfactory performance has been plotted only for samples or equipment exposed for at least one year. Unsatisfactory performance from laboratory or plant data has been plotted regardless of the length of exposure time. The chemical compositions of the steels in Figure 1 should conform to the limits specified for the various grades by ASTM/ASME.

Since the original version of Figure 1 was prepared for API in 1949,[4] further experience has enabled curves for most commonly used steels to be more accurately located. A major exception has been for C–0.5Mo steel. This edition consolidates all information relevant to 0.5Mo steel (including C–0.5Mo) in Appendix A.

This edition has also added three data points which show high temperature hydrogen attack of 1.25Cr–0.5Mo steel below the current 1.25Cr–0.5Mo curve. See Appendix B for more discussion of 1.25Cr–0.5Mo steel. Appendix C gives a similar discussion for 2.25Cr–1.0Mo steel.

## 3.2 SELECTING MATERIALS FOR NEW EQUIPMENT

Figure 1 is often used when selecting materials for new equipment in hydrogen service. When using Figure 1 as an aid for materials selection, it is important to recognize that Figure 1 only addresses a material's resistance to high temperature hydrogen attack. It does not take into account other factors important at high temperatures, such as:

a. Other corrosives that may be in the system, such as hydrogen sulfide.
b. Creep, temper embrittlement, or other high temperature damage mechanisms.
c. Possible synergistic effects, such as between high temperature hydrogen attack and creep.

Temperatures for data plotted in the figures represent a range in operating conditions of ±20°F (±11°C). Because Figure 1 is based largely upon empirical experience, an operating company may choose to add a safety margin, below the relevant curve, when selecting steels.

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## References and Comments for Figure 1

The data points in Figure 1 are labeled with reference numbers corresponding to the sources listed below. The letters in the figure correspond to the comments listed on this page.

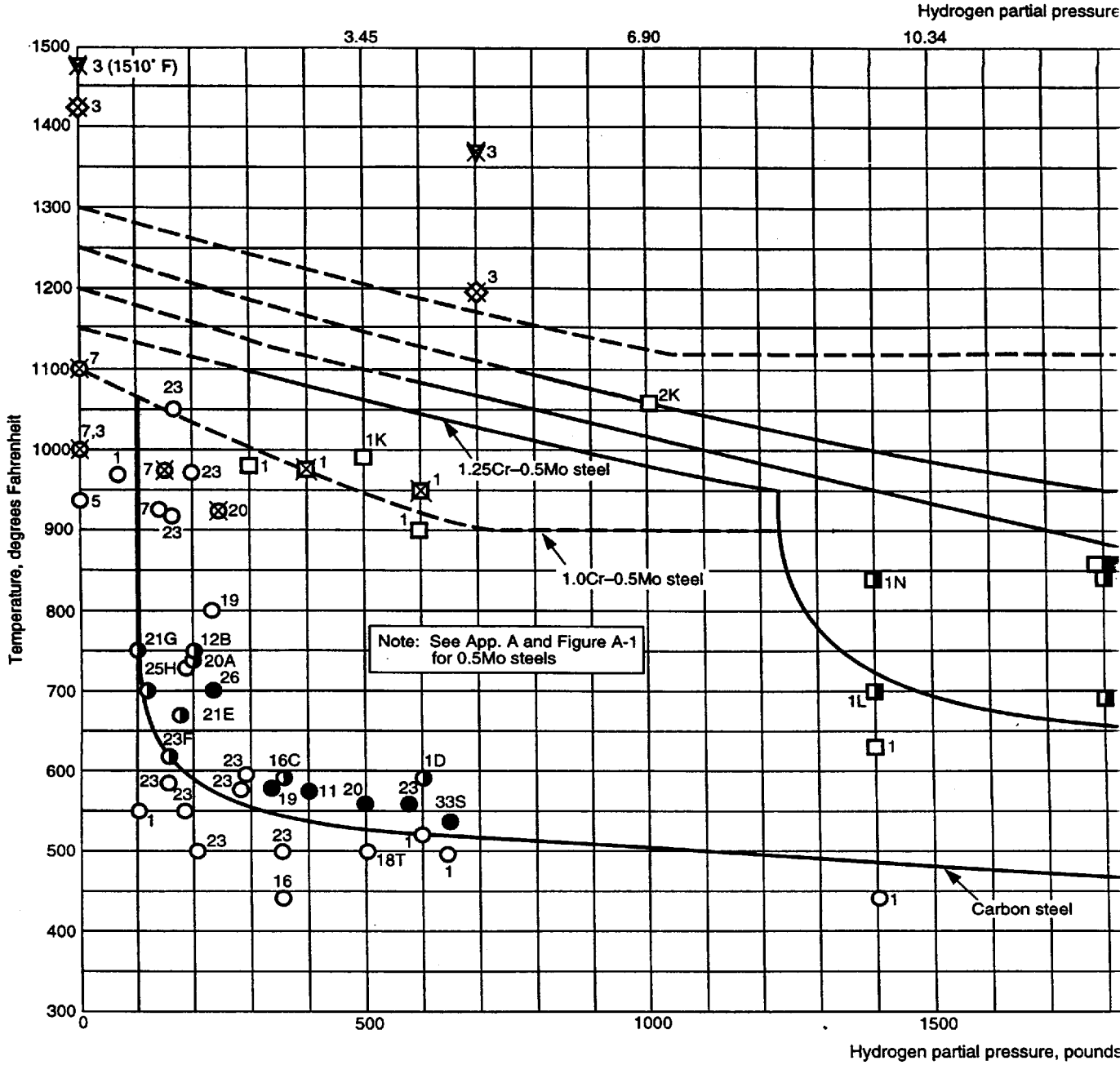
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### Comments

A. A section made of A 106 pipe was found to be attacked to 27 percent of its thickness after 5745 hours. Other pieces of pipe in the same line were unaffected.
B. The attack was concentrated in the overheated section of a hot bent steel elbow. The unheated straight portions of the elbow were not attacked.
C. In a series of 29 steel samples, 12 were attacked while 17 were not.
D. After 2 years' exposure, five out of six pieces of carbon steel pipe were attacked. One piece of pipe was unaffected.
E. Attack was concentrated in the weld and heat-affected sections of A 106 pipe. Metal on either side of this zone was unaffected.
F. After 11 years' service, attack was found in the hot bent section of A 106 pipe. Unheated straight sections were not affected.
G. After 2 years' service, all parts of carbon steel pipe, including weld and heat-affected zones, were satisfactory.
H. After 4 years' service, weld and heat-affected zones of A 106 pipe showed cracks.
J. After 31 years' service, a forging of 0.3C–1.3Cr–0.25Mo steel showed cracks 0.007 inch (0.2 millimeter) deep.
K. Pipes of 1.25Cr-0.25Mo steel.
L. After 4 years' service, a forging of 0.3C–1.3Cr–0.25Mo steel was unaffected.
N. After 7 years' service, a forging of 0.3C–1.52Cr–0.50Mo steel showed cracks 0.050 inch (1.3 millimeters) deep.
P. After 30 years' service, a forging of 0.30C–0.74Cr–0.43Ni steel was unaffected.
Q. After 15 years in ammonia service, a pipe of 0.15C–2.25Cr–1.00Mo steel showed no HTHA but was nitrided to a depth of 0.012 inch (0.3 millimeter).
S. After 8 years, carbon steel cracked.
T. After 18 years, carbon steel did not show HTHA.
U. After 450 days' exposure, 1.25Cr–0.5Mo valve body was not damaged by HTHA.

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Notes:

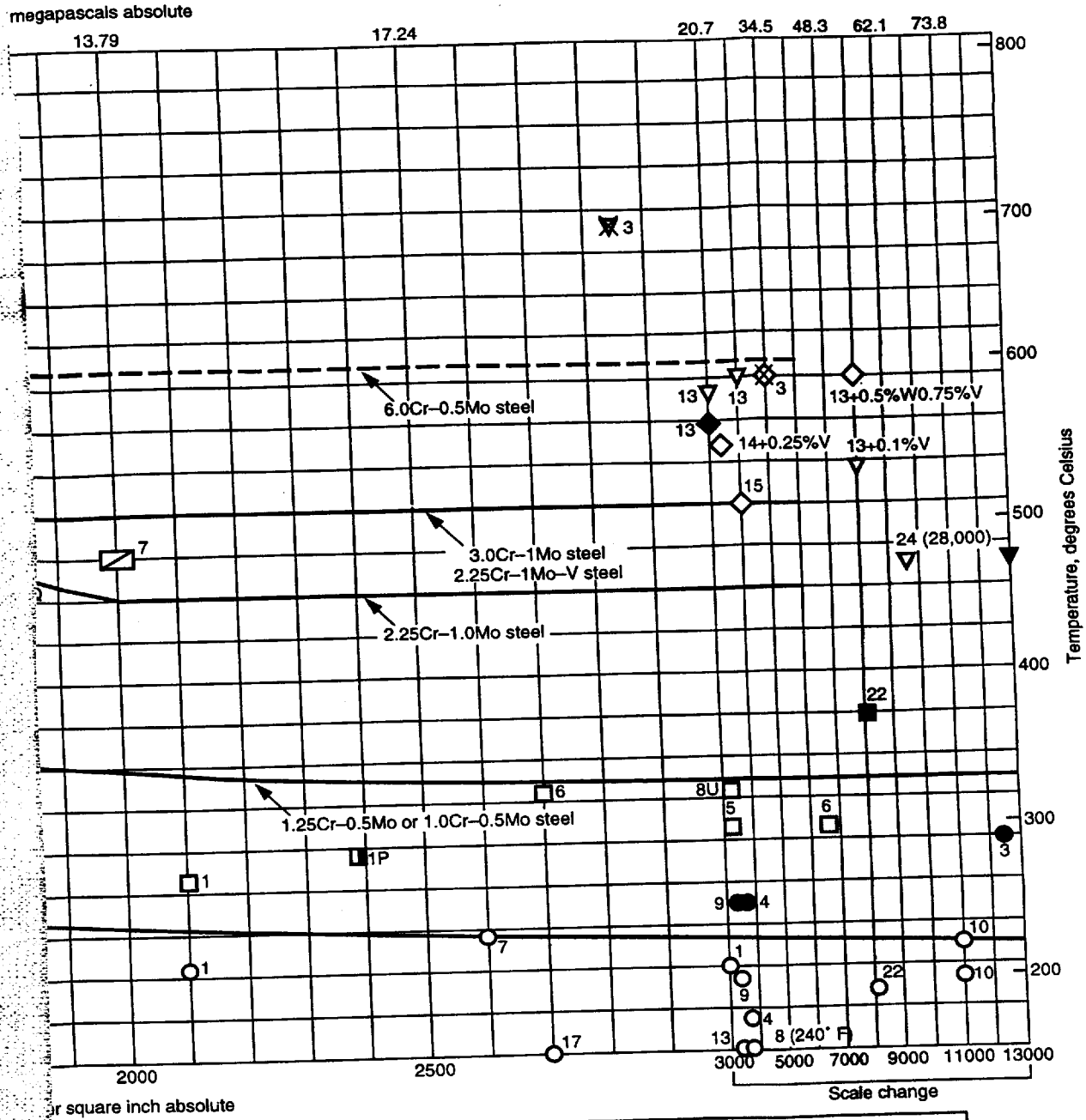
1. The limits described by these curves are based on service experience originally collected by G.A. Nelson and on additional information gathered by or made available to API.
2. Austenitic stainless steels are generally not decarburized in hydrogen at any temperature or hydrogen pressure.
3. The limits described by these curves are based on experience with cast steel as well as annelaed and normalized steels at stress levels defined by Section VIII, Division I, of the ASME code. See 5.3 and 5.4 in text for additional information.
4. Several failures of 1-1/4Cr–1Mo steel have been reported in the satisfactory region. See Appendix B for details.
5. The inclusion of the 2.25Cr–1Mo-V class of steels is based on 10,000+hr laboratory tests where these alloys were at least equal to the 3Cr–1Mo steel. See Reference 22 listed in Section 2.2.

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This figure was revised by API in 1969, 1983, 1990 and 1996.

API 941 Fifth Edition
January 1997

Figure 1—Operating Limits for Steels in Hydrogen S

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Legend:

Surface decarburization

Internal decarburization

| | Carbon steel | 1.0Cr 0.5Mo | 2.25Cr 1.00Mo | 3.0Cr 1.0Mo | 6.0Cr 0.5Mo |
|---|---|---|---|---|---|
| Satisfactory | ○ | □ | ▭ | ◇ | ▽ |
| Internal decarburization and fissuring | ● | ■ | ▬ | ◆ | ▼ |
| Surface decarburization | ⊠ | ⊠ | ▨ | ⊛ | ⋇ |
| See comments | ◑ | ◧ | ◨ | ◈ | ▼ |

vice to Avoid Decarburization and Fissuring

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# 4 Forms of High Temperature Hydrogen Attack

## 4.1 GENERAL

As noted above, high temperature hydrogen can attack steels in two ways:

a. Surface decarburization.
b. Internal decarburization and fissuring.

The combination of high temperature and low hydrogen partial pressure favors surface decarburization without internal decarburization and fissuring. The combination of low temperature, but above 430°F (221°C), and high hydrogen partial pressure favors internal decarburization and fissuring. These mechanisms are described more fully below.

The broken-line curves at the top of Figure 1 represent the tendencies for surface decarburization of steels while they are in contact with hydrogen. The solid-line curves represent the tendencies for steels to decarburize internally with resultant fissuring and cracking created by methane formation.

## 4.2 SURFACE DECARBURIZATION

Surface decarburization does not produce fissures. In this respect, it is similar to decarburization created by the exposure of steel to certain other gases, such as air, oxygen, or carbon dioxide. The usual effects of surface decarburization are a slight, localized reduction in strength and hardness and an increase in ductility. Because these effects are usually small, there is often much less concern with surface decarburization than there is with internal decarburization.

A number of theories have been proposed to explain this phenomenon,[5,6,7] but the currently-accepted view is based on the migration of carbon to the surface where gaseous compounds of carbon are formed, rendering the steel less rich in carbon. (The gaseous compounds formed are $CH_4$ or, when oxygen-containing gases are present, CO.) Water vapor hastens the reaction. Carbon in solution diffuses to the surface so that the rate-controlling mechanism appears to be carbon diffusion. Inasmuch as the carbon in solution is continuously supplied from the carbides, carbide stability is directly related to the rate of surface decarburization.

In cases where surface decarburization predominates over internal attack, the actual values of pressure-temperature combinations have not been extensively studied; but the limits defined by Naumann[8] probably give the most accurate trends. Naumann's work, which is based on 100-hour tests, indicates decarburization tendencies; however, long-time exposures have indicated lower operating limits.

## 4.3 INTERNAL DECARBURIZATION AND FISSURING

The solid-line curves in Figure 1 define the areas above which material damage by internal decarburization and fissuring have been reported. Below and to the left of the curve for each alloy, satisfactory performance has been experienced with periods of exposure of up to approximately 40 years. At temperatures above and to the right of the solid curves, internal decarburization occurs. Internal decarburization and fissuring are preceded by an incubation period that depends on temperature and hydrogen partial pressure (see Section 5.2 for further discussion).

Internal decarburization and fissuring are caused by hydrogen permeating the steel and reacting to form other gases, such as methane.[8] The methane formed cannot diffuse out of the steel and typically accumulates at grain boundary voids. This results in high localized stresses which lead to the formation of fissures, cracks, or blisters in the steel. Fissures in hydrogen-damaged steel lead to a substantial deterioration of mechanical properties.

Figure 2 shows the microstructure of a sample of C–0.5Mo steel damaged by internal decarburization and fissuring. The service conditions were 790°F (421°C) at a hydrogen partial pressure of 425 pounds per square inch absolute (2.9 megapascals) for approximately 65,000 hours in a catalytic reformer.

The addition of carbide stabilizers to steel reduces the tendency toward internal fissuring. Elements such as chromium, molybdenum, tungsten, vanadium, titanium, and niobium reduce the number of nucleation sites by forming more stable alloy carbides which resist breakdown by hydrogen and, therefore, decrease the propensity to form methane.[9] The solid-line curves in Figure 1 reflect the increased resistance to internal attack when molybdenum and chromium are present.

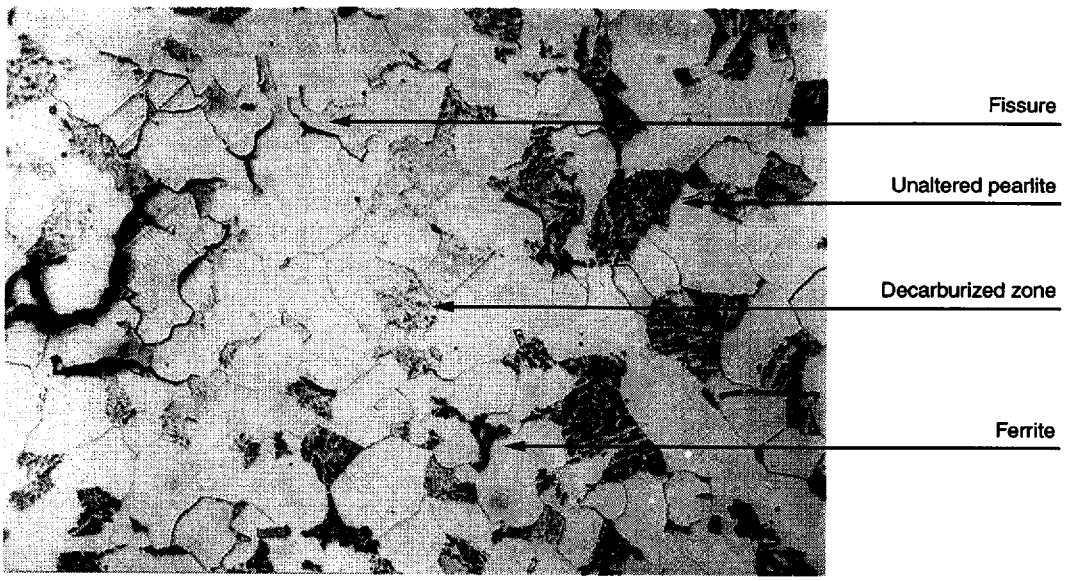
The presence of nonmetallic inclusions tends to increase the extent of blistering damage. When steel contains segregated impurities, stringer-type inclusions or laminations, hydrogen or methane accumulations in these areas may cause severe blistering.[10]

Alloys other than those shown in Figure 1 are also suitable for resisting high temperature hydrogen attack. These include modified carbon steels and low alloy steels to which carbide stabilizing elements (molybdenum, chromium, vanadium, titanium, or niobium) have been added. European alloys and heat-treating practices have been summarized by Class.[11] Austenitic stainless steels are resistant to decarburization even at temperatures above 1000°F (538°C).[12]

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Notes:
1. Service conditions were 65,000 hours in a catalytic reformer at a temperature of 790°F (421°C) and a hydrogen partial pressure of 425 pounds per square inch absolute (2.9 megapascals). From Reference 14.
2. Magnification: 520×; nital etched.

Figure 2—C–0.5Mo Steel (ASTM A 204-A) Showing Internal Decarburization and Fissuring In High Temperature Hydrogen Service

## 5 Factors Influencing HTHA

### 5.1 HIGH TEMPERATURE HYDROGEN ATTACK IN A LIQUID HYDROCARBON PHASE

High temperature hydrogen attack can occur in a liquid hydrocarbon phase if it can occur in the gas phase in equilibrium with the liquid phase. For materials selection purposes (using Figure 1), hydrogen dissolved in liquid hydrocarbon should be assumed to exert a vapor pressure equal to the hydrogen partial pressure of the gas with which the liquid is in equilibrium. Recent plant experience and testing of field-exposed specimens have shown that high temperature hydrogen attack can occur under such conditions.[13]

High temperature hydrogen attack has been found in carbon steel, liquid-filled piping downstream of a heavy oil desulfurization unit separator that was operating at hydrogen partial pressure and temperature conditions above the Figure 1 carbon steel curve. Testing of field-exposed test specimens showed high temperature hydrogen attack of both chrome-plated and bare carbon steel samples which were totally immersed in liquid.[13]

### 5.2 INCUBATION

Damage to steels by high pressure, high temperature hydrogen is preceded by a period of time when no noticeable change in properties is detectable by current mechanical testing methods. After this period of time has elapsed, material damage is evident with resultant decreases in strength, ductility, and toughness. The length of time before high temperature hydrogen attack can be detected by usual mechanical testing methods is termed the incubation period. This period varies with the type of steel and severity of exposure; it may last only a few hours under extreme conditions and become progressively longer at lower temperatures and hydrogen partial pressures. With some steels under mild conditions, no damage can be detected by mechanical testing methods even after many years of exposure.

The length of the incubation period is important because it determines the useful life of a steel at conditions under which high temperature hydrogen attack occurs. Useful theoretical models of the high temperature hydrogen attack mechanism and incubation period have been proposed.[14, 15, 16] High temperature hydrogen attack can be viewed as occurring in three stages:

a. The incubation period during which mechanical properties change very slowly.
b. The stage of rapid mechanical property deterioration associated with rapid fissure growth.
c. The final stage where carbon in solid solution is exhausted and mechanical properties reach their final value.

During the incubation period, methane pressure builds up in submicroscopic voids. These voids grow slowly due both to internal methane pressure and applied stress. When the

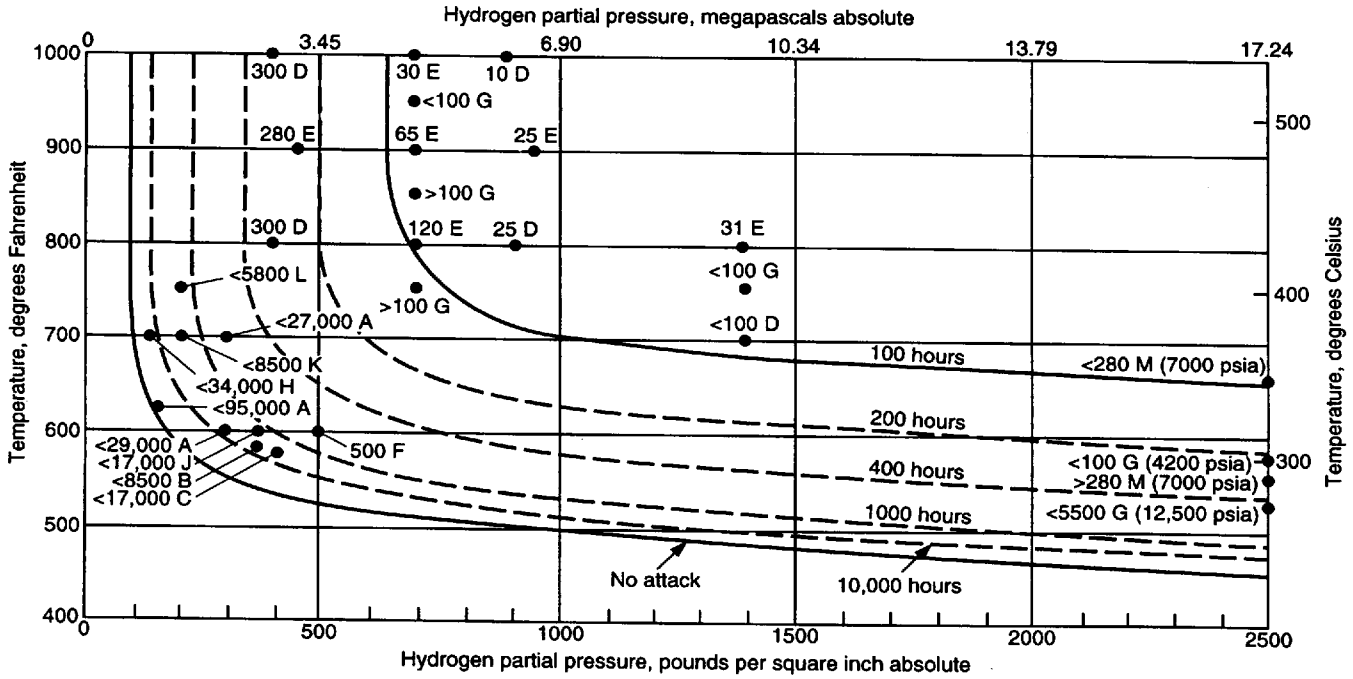
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voids reach a critical size, the effects on mechanical properties become significant. The incubation period depends on many variables, including the type of steel, degree of cold working, amount of impurity elements, applied stress, hydrogen pressure, and temperature.

Incubation curves for carbon steel are given in Figure 3.[17] These can be used as a guide in determining approximate safe operating times for steels that operate above their long-term experience curves. Appendix A includes similar curves that may be useful for some heats of C–0.5Mo steel, with the precaution that resistance of C–0.5Mo steel to high temperature hydrogen attack is particularly sensitive to heat treatment, chemical composition, and the heating/cooling history of the steel during forming.[18, 19, 20, 21]

## 5.3 EFFECT OF PRIMARY STRESSES

Many users have reported satisfactory performance of annealed or normalized and tempered steels produced before 1969, as shown in Figure 1. These steels have been used for pressure-retaining equipment at design stress levels allowed by the 1969 or earlier editions of commonly-accepted codes (such codes include the ASME Code, Section VIII, Division 1; the standards of the American National Standards Institute; and, for the lower-strength materials, those of Deutsche Industrie-Normen). However, pressure vessels in hydrogen service have also been constructed using the higher allowable stresses permitted in either Section VIII, Division 2, or modifications of Section III of the ASME Code. Quenched and tempered or normalized and tempered steels have normally




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Figure 3—Time For Incipient Attack of Carbon Steel in High Temperature Hydrogen Service

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been utilized for these vessels due to their improved mechanical properties (strength and impact toughness).

No incidents of decarburization or fissuring of pressure vessels built to the specifications of Section VIII, Division 2, of the ASME Code have been reported. None of the failure data in Figure 1 represent materials used at the higher allowable stresses.

Recently published laboratory creep studies[22] have shown that the rupture strength and rupture ductility of 2.25Cr–1Mo steel are diminished when tested in hydrogen as compared to their values in air. These tests were a continuation of previously-reported tests[23, 24, 25] that showed somewhat conflicting results in shorter term tests.

These recent tests were conducted at applied stress levels similar to those that might be experienced by ASME Section VIII, Division 2 vessels. Test exposure times exceeded 50,000 hours depending on applied stress and temperature. The test specimens were from weldments of thick section plates and represented base metal, weld metal, and heat-affected zone. Detrimental effects of hydrogen were found down to the Figure 1 limit of 850°F (454°C) at 2000 pounds per square inch absolute (14 megapascals) and 3000 pounds per square inch absolute (21 megapascals) hydrogen partial pressure.

## 5.4 EFFECT OF SECONDARY STRESSES

High temperature hydrogen attack can be accelerated by secondary stresses, such as thermal stresses or those induced by cold work. High thermal stresses were considered to play a significant role in the high temperature hydrogen attack of some 2.25Cr–1Mo steel piping.[26] Other 2.25Cr–1Mo steel piping in the same system, subjected to more severe hydrogen partial pressures and temperatures, was not attacked.

The effect of cold work was demonstrated by Vitovec in work sponsored by API and summarized in API Publication 940.[9] Vitovec compared specific gravities of SAE 1020 steel with varying degrees of cold work tested in 900 psi (6.2 megapascals) hydrogen at 700°F (371°C), 800°F (427°C), and 1000°F (538°C). The decrease in specific gravity over time indicates the rate at which internal fissures produced by high temperature hydrogen attack are developed.

Annealed samples (0 percent strain) had an incubation period followed by a decrease in specific gravity. Steels with 5 percent strain had shorter incubation periods, and specific gravity decreased at a more rapid rate. Steels with 39 percent strain showed no incubation period at any test temperature, indicating that fissuring and cracking started immediately upon exposure to hydrogen.

These tests are considered significant in explaining the cracks sometimes found in highly stressed areas of an otherwise apparently resistant material. In addition, Cherrington and Ciuffreda[27] have emphasized the need for removing notches (stress concentrators) in hydrogen service equipment.

## 5.5 EFFECT OF HEAT TREATMENT

Both industry experience and research work indicate that postweld heat treatment (PWHT) of chromium-molybdenum steels in hydrogen service improves resistance to high temperature hydrogen attack. The PWHT stabilizes alloy carbides. This reduces the amount of carbon available to combine with hydrogen, thus improving high temperature hydrogen attack resistance.

Research[7,16,20, 21, 28] has shown that certain metal carbides may be more resistant to decomposition in high temperature hydrogen environments. Creep tests in hydrogen demonstrated the beneficial effect of increased PWHT on the high temperature hydrogen attack resistance of 2.25Cr–1Mo steel.[22] In these tests, 2.25Cr–1Mo steels postweld heat treated for 16 hours at 1275°F (691°C) showed more resistance to high temperature hydrogen attack than the same steels postweld heat treated for 24 hours at 1165°F (630°C). Both high PWHT temperatures and longer times are beneficial. Similarly, high temperature hydrogen attack resistance of 1Cr–0.5Mo and 1.25Cr–0.5Mo steels is improved by raising the minimum PWHT temperature to 1250°F (677°C) from the 1100°F (593°C) minimum required by Section VIII of the ASME Code.

The user must balance the advantages of high PWHT temperatures with other factors, such as the effect upon strength and notch toughness. Note higher PWHT temperatures can affect the ability to meet ASME Code Class 2 strength requirements.

## 5.6 EFFECT OF STAINLESS STEEL CLADDING OR WELD OVERLAY

The solubility of hydrogen in austenitic stainless steel is about an order of magnitude greater than for ferritic steels.[29] The diffusion coefficient of hydrogen through austenitic stainless steel is roughly two orders of magnitude lower than for ferritic steels.[30, 31] A sound, metallurgically bonded austenitic stainless steel cladding or weld overlay can reduce the effective hydrogen partial pressure acting on the base metal. Ferritic or martensitic claddings or weld overlays have little or no benefit in reducing the hydrogen partial pressure acting on the base metal.

The amount of hydrogen partial pressure reduction depends upon the materials and the relative thickness of the cladding/weld overlay and the base metal—the thicker the stainless barrier is relative to the base metal the better.[32] Archakov and Grebeshkova[33] mathematically considered how stainless steel corrosion barrier layers increase resistance of carbon and low alloy steels to high temperature hydrogen attack.

There have been a few instances of high temperature hydrogen attack of base metal which was clad or overlayed with austenitic stainless steel. All of the reported instances involved C–0.5Mo steel base metal. In one case,[34] high tem-

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perature hydrogen attack occurred in a reactor vessel at a nozzle location where the C–0.5Mo base metal was very thick relative to the cladding/overlay. Another incident of high temperature hydrogen attack of C–0.5Mo steel occurred under intergranularly cracked Type–304 austenitic stainless steel cladding (see data point 51U in Appendix A). The other cases involved ferritic or martensitic stainless steel cladding.

It is not advisable to take a credit for the presence of an austenitic stainless steel cladding/weld overlay when selecting the base metal for a new vessel or when operating an existing vessel long term in high temperature hydrogen service. In practice, satisfactory performance of a lower alloy base metal requires assurance that the effective hydrogen partial pressure acting on the base metal be accurately determined and that the integrity of the cladding/weld overlay be maintained. Such assurance may be difficult to achieve, especially where complex geometries are involved. In a very few instances operators have taken some credit for the presence of an austenitic stainless steel cladding/weld overlay for short-term operation when conditions only marginally exceeded the Figure 1 curve for the base metal. Many operators, however, take the presence of an austenitic stainless steel cladding/weld overlay into account when establishing inspection priorities for HTHA, especially for C–0.5Mo steel equipment.

# 6 Inspection

## 6.1 INSPECTION FOR HIGH TEMPERATURE HYDROGEN ATTACK

The selection of optimum inspection methods and frequencies for high temperature hydrogen attack in specific equipment or applications is the responsibility of the user. The information below and in Tables D-1 and D-2 are intended to assist the user in making such decisions.

Experience with steels operated below their respective Figure 1 Nelson curves has been good. Consequently, most users do not inspect equipment for high temperature hydrogen attack damage unless it has been operated near or above its curve. Only a few documented instances of high temperature hydrogen attack occurring at conditions below the curves have been reported to the API (see Appendixes A, B, and C). C–0.5Mo steel is the notable exception to this[35] and periodic inspection should be considered if operated above the carbon steel curve, based on factors such as relative position of the operating parameters versus the carbon steel curve, consequence of failure, presence of cladding, prior heat treatment, etc. See also Appendix A.

High temperature hydrogen attack damage may occur in welds, weld HAZs, or base metal. Even within these specific areas, the degree of damage may vary widely. Consequently, if damage is suspected, then a thorough inspection means that representative samples of these areas be examined.

Tables D-1 and D-2 provide a summary of available methods of inspection for HTHA damage, including a discussion of the advantages and limitations of each. Two or more inspection methods are often used in combination to overcome the limitations of any single method.[36]

High temperature hydrogen attack is a difficult inspection challenge. The early stages of attack with microfissures, or even small cracks, can be difficult to detect. The advanced stage of attack with significant cracking is much easier to detect, but at that point there is already a risk of equipment failure. In addition to general attack of the base metal, high temperature hydrogen attack has been known to occur as a very narrow band of intense attack and cracking, running alongside and parallel to welds. This highly localized form of attack is especially difficult to detect.

For base metal examination, ultrasonic methods are the most effective in early detection of high temperature hydrogen attack damage in the microfissure stage. Most effective is the use of a sensitive backscatter method in combination with the velocity ratio, attenuation, and/or spectral analysis techniques. Backscatter can be used as a first step of inspection and can be used to quantify the depth of damage. Velocity ratio and attenuation are useful for confirmation of backscatter indications. Other methods are capable of detecting high temperature hydrogen attack only after discrete cracks have formed and there is significant degradation of mechanical properties.

For weldment examination, there is no published nondestructive examination method (other than possibly in situ metallography, see below) that will reliably detect high temperature hydrogen attack damage in the microfissure stage. The nature of weld grain structures makes ultrasonic examination difficult. Wet fluorescent magnetic particle inspection can effectively detect only surface-breaking cracks in most cases, and access to the internal surfaces of the base metal of the equipment is required. When such access is not possible, and for clad/overlay equipment and most piping, ultrasonic shear wave inspection from the external surface may be used.

In situ metallography can be effective in detecting the early stages of high temperature hydrogen attack at the surface of the steel. Skill is required for the surface polishing, etching, replication, and microstructural interpretation. Because in situ metallography only examines a small specific area, other methods should be used to complement it. It requires access to the surface of interest, and may require removal of a small amount of surface material from the process side for best results (see Table D-2).

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# APPENDIX A—HIGH TEMPERATURE HYDROGEN ATTACK OF 0.5Mo STEELS

The purpose of this appendix is to provide a brief summary of the information and experience regarding the use of 0.5Mo (C–0.5Mo, Mn–0.5Mo) steels in elevated temperature and pressure hydrogen service.

Most companies no longer specify C–0.5Mo steel for new or replacement equipment used for operation above the carbon steel curve in Figure 1 because of the uncertainties regarding its performance after prolonged use. Since 1970, a series of unfavorable service experiences with C–0.5Mo steels has reduced confidence in the position of the 0.5Mo curve.[A1, A2] In the second edition (1977) of this publication, the 0.5Mo curve was lowered approximately 60°F (33°C) to reflect a number of plant experiences that involved high temperature hydrogen attack of C–0.5Mo equipment. In the fourth edition (1990) of this publication, the 0.5Mo curve was removed from Figure 1 due to additional cases of high temperature hydrogen attack of C–0.5Mo steel equipment as much as 200°F (111°C) below the curve. Plant experience has identified 27 instances of high temperature hydrogen attack below the 1977 curve. The operating conditions for these instances are given in Table A-1, and are plotted on Figure A-1.

No instances have been reported of high temperature hydrogen attack of Mn–0.5Mo steel below the Figure A-1 0.5Mo curve. The information and use of this material at elevated temperatures and hydrogen partial pressures are limited.

C–0.5Mo steels vary in their resistance to high temperature hydrogen attack. Many heats seem to have resistance at conditions indicated by the 0.5Mo curve on Figure A-1. However, some heats seem to have high temperature hydrogen attack resistance only marginally better than carbon steel. Recent work[A2,A3,A4,A5] relating its resistance to high temperature hydrogen attack to the thermal history of the steel give some guidance as to why. Slow-cooled, annealed C–0.5Mo steels have less resistance to high temperature hydrogen attack than normalized steels. The studies have shown that postweld heat treatment improves the high temperature hydrogen attack resistance of weldments and heat affected zones for both annealed and normalized C–0.5Mo steels. However, the base metals of slow-cooled, annealed C–0.5Mo steels show a decrease in high temperature hydrogen attack resistance after postweld heat treatment. The initial studies suggest that this is due to free carbon being present in the ferrite matrix after postweld heat treatment. Normalized C–0.5Mo steel base metals, on the other hand, show improvement in high temperature hydrogen attack resistance following tempering or postweld heat treatment. Such normalized and postweld heat treated C–0.5Mo steel appears to have hydrogen attack resistance about as indicated by the 0.5Mo curve in the second edition (1977) of this publication. Until the factors controlling the high temperature hydrogen attack resistance of C–0.5Mo are better understood, each user should carefully assess the use of C–0.5Mo steel in services above the carbon steel curve in Figure A-1.

Existing C–0.5Mo steel equipment that is operated above the carbon steel curve in Figure A-1 should be inspected to detect high temperature hydrogen attack. Owners/operators should evaluate and prioritize C–0.5Mo equipment operating above the carbon steel limit for inspection—Hattori and Aikawa[A6] addressed this issue. The work cited above and plant experience suggest that important variables to consider in prioritizing equipment for inspection include severity of operating condition (hydrogen partial pressure and temperature), thermal history of the steel during fabrication, stress, cold work, and cladding composition and thickness, when present.

To provide a historical summary of the data regarding the use of C–0.5Mo steels, two additional figures are included here: (1) Figure A-2 which shows the effect of trace alloying elements (molybdenum) on operating limits, and (2) Figure A-3 which shows incubation times for C–0.5Mo steels. Figure A-2 is from the second edition of this publication (1977), and is a revision of a similar figure from the original edition (1970). Figure A-2 shows that molybdenum has long been considered to be beneficial to the high temperature hydrogen attack resistance of steels. The data in Figures A-2 and A-3 should be used with caution, since some heats of C–0.5Mo steels have suffered high temperature hydrogen attack during exposure to conditions under the lower solid curve (equivalent to the C–0.5Mo curve of Figure A-1). The data for the instances of high temperature hydrogen attack listed in Table A-1 and plotted on Figure A-1 are also shown for reference in Figure A-3. The incubation time would be less than the service life at the time the attack was detected.

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## REFERENCES AND COMMENTS FOR FIGURE A-1

The data in Figure A-1 are labeled with reference numbers corresponding to the sources listed below. The letters in the figure correspond to the comments listed on this page.

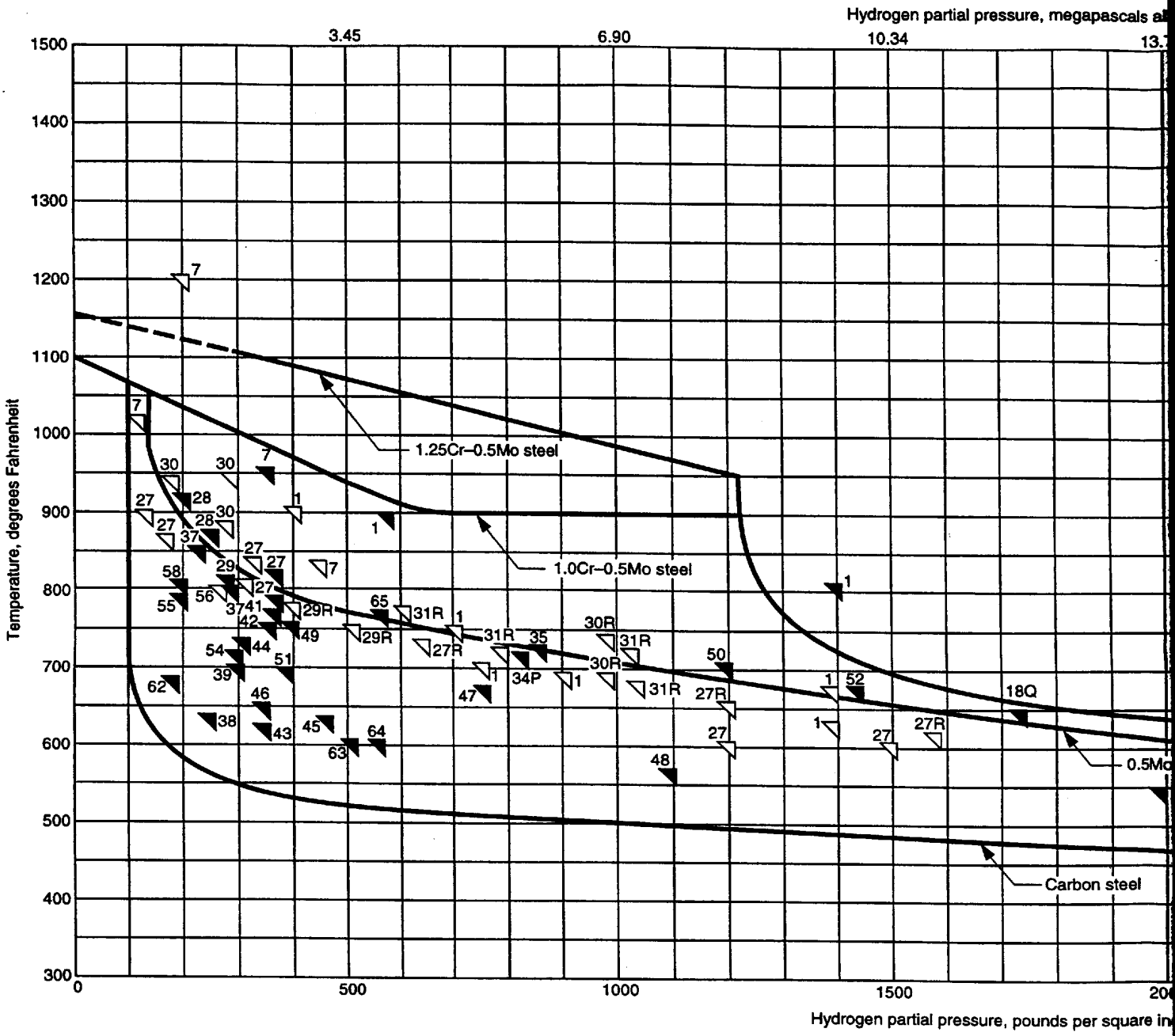
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### Comments

A. Feed line pipe leaked; isolated areas damaged. Blistered, decarburized, fissured; postweld heat treated at 1100°F to 1350°F.
B. Effluent line, pipe and heat-affected zone, isolated areas damaged; no postweld heat treatment.
C. Weld and pipe, isolated areas damaged; no postweld heat treatment.
D. Effluent line; weld, isolated areas damaged; postweld heat treatment.
E. Feed line; weld and heat-affected zone, isolated areas damaged; postweld heat treatment.
F. Feed/effluent exchanger nozzle-to-shell weld, cracks in welds and in exchanger tubes.
G. Effluent exchanger channel; welds, plate, and heat-affected zone, isolated areas damaged; postweld heat treatment.
H. Effluent exchanger channel; welds, plate, and heat-affected zone, isolated areas damaged; postweld heat treated at 1100°F.
I. Catalytic reformer, combined feed/effluent exchanger shell; plate; postweld heat treated at 1250°F.
J. Hydrodesulfurization unit effluent exchanger channel head and shell plate. (Hydrocarbon feed to unit and make-up hydrogen from ethylene unit.)
K. Catalytic reformer combined feed piping; welds and base metal; postweld heat treatment.
L. Gas-oil hydrodesulfurization unit. Elbow cracked intergranularly and decarburized at fusion line between weld metal and heat-affected zone; no postweld heat treatment.
M. Ammonia plant converter; exit piping; intergranular cracking and internal decarburization of pipe.
P. Hydrodesulfurization unit hydrogen preheat exchanger shell; blisters, intergranular fissuring, and decarburization in weld metal; postweld heat treated at 1150°F.
Q. Attack of heat exchanger tubing in tubesheet.
R. Stainless steel cladding on 0.5Mo steel; no known HTHA.
S. Decarburization and fissuring of weld metal; postweld heat treated at 1150°F.
T. Forged tubesheet cracked with surface decarburation; tubes blistered.
U. Hydrodesulfurization unit, C-0.5Mo steel exchanger tubesheet; decarburized, fissured, and cracked under intergranularly cracked ASTM Type 304 cladding.
V. Hydrocracker charge exchanger liquid with a small amount of hydrogen; C-0.5Mo with Type 410S rolled bond clad. Extensive blistering and fissuring under clad.
W. C-0.5Mo steel piping in ammonia plant syngas loop; decarburized and fissured.
AA. Blistering and fissuring of a flange.
BB. HAZ and base metal fissuring of pipe.
CC. Base metal fissuring and surface blistering in heat exchanger shell.
DD. Attack at weld, HAZ and base material in piping.
EE. Localized attack in weld, HAZ in piping.
FF. Base metal attack in piping.
GG. Base metal attack in a heat exchanger channel.
HH. Base metal attack in piping.
II. Blistering and base metal attack in a heat exchanger shell.
JJ. Base metal attack in a TP405 roll bond clad vessel.
KK. Base metal attack in a TP405 roll bond clad vessel.
LL. Attack in nozzle attachment area of a vessel weld overlaid with Type 309Nb.
MM. Internal decarburization/fissuring of piping in a hydrocracker unit after 235,000 hours of service.

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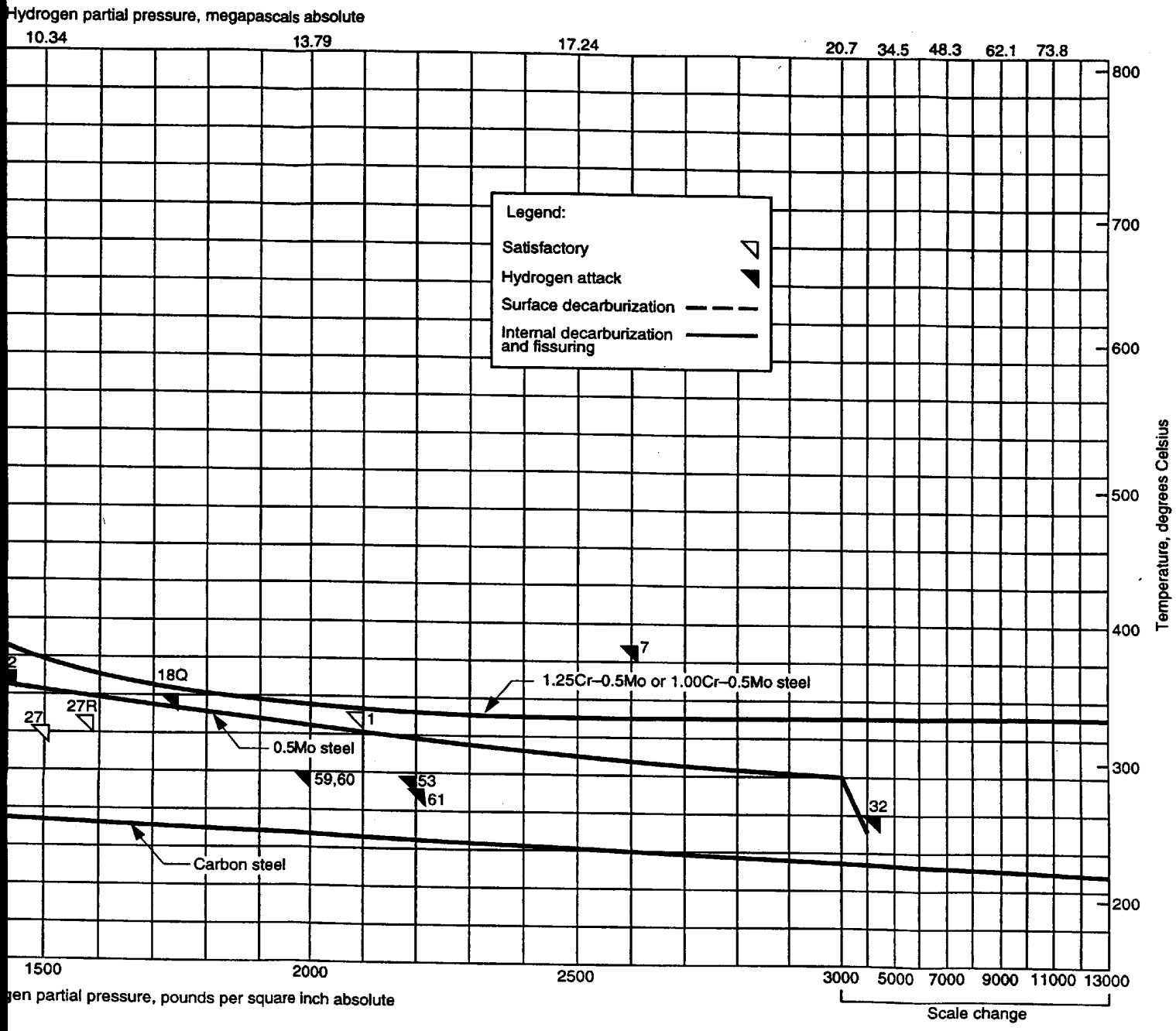

Notes:
1. References and comments are shown on Table A-1.
2. Curves for carbon steel, 1.0Cr–0.5Mo steel, and 1.25Cr–0.5Mo steel are included for reference.
3. The symbol ◹ is retained as a reference against previous revisions of this publication.
4. Reference numbers are the same as in previous editions of this publication.
5. The 0.5Mo steel curve is the same as the one shown in the fourth edition of this publication (1990).

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Figure A-1—Experience with C–0.5Mo and Mn–0.5Mo Steel in Hi

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–0.5Mo and Mn–0.5Mo Steel in High Temperature Hydrogen Service

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Table A-1—Operating Conditions For C–0.5Mo Steels That Experienced High Temperature Hydrogen Attack Below the 0.5Mo Steel Curve in Figure A-1

| Point | Temperature | | Hydrogen Partial Pressure (Absolute) | | Service Hours (Approx.) | Degrees Below 0.5Mo Curve (Approximate) | |
|---|---|---|---|---|---|---|---|
| | °F | °C | psi | MPa | | °F | °C |
| 36A[1] | 790 | 421 | 350 | 2.41 | 80,000 | 20 | 11 |
| 37B[1] | 800 | 427 | 285 | 1.97 | 57,000 | 30 | 17 |
| 38C[1] | 640 | 338 | 270 | 1.86 | 83,000 | 180 | 100 |
| 39D[1] | 700 | 371 | 300 | 2.07 | 96,000 | 125 | 69 |
| 41F[1] | 760 | 404 | 375 | 2.59 | 85,000 | 40 | 22 |
| 42G[1] | 750 | 399 | 350 | 2.41 | 150,000 | 60 | 33 |
| 43H[1] | 625 | 329 | 350 | 2.41 | 150,000 | 185 | 103 |
| 44I[1] | 730[2] | 388[2] | 313 | 2.16 | 116,000 | 90 | 50 |
| 45J[3] | 620[2]/640[4] | 327/338 | 457 | 3.15 | 70,000 | 167/147 | 93/82 |
| 46K[1] | 626[2]/680[4] | 330/360 | 350 | 2.41 | 131,000 | 184/130 | 102/72 |
| 47L[3] | 684[2] | 362 | 738 | 5.09 | 61,000 | 54 | 30 |
| 48M[5] | 550/570 | 288/299 | 1060/1100 | 7.31/7.59 | 79,000 | 125/105 | 69/58 |
| * | 655/670 | 346/354 | - | - | 17,500 | 20/5 | 11/3 |
| 49S[3] | 750/770 | 399/410 | 390 | 2.69 | 67,000 | 50/30 | 28/17 |
| * | 650 | 343 | - | - | 163,000 | 150 | 83 |
| 51U[3] | 690 | 366 | 397 | 2.74 | - | 100 | 56 |
| 53W[5] | 545 | 285 | 2190 | 15.1 | 140,000 | 45 | 25 |
| 54AA[1] | 725[2]/760[4] | 385[2]/404[4] | 300[2]/380[4] | 2.07[2]/2.62[4] | 105,000 | 40/100 | 22/56 |
| 55BB[1] | 800[2]/850[4] | 427[2]/454[4] | 175[2]/190[4] | 1.21[2]/1.31[4] | 124,000 | 80/30 | 44/17 |
| 56CC[1] | 810[2]/825[4] | 432[2]/441[4] | 275[2]/300[4] | 1.90[2]/2.07[4] | 223,000 | 15/0 | 8/0 |
| 57DD[1] | 850[4] | 454[4] | 225[4] | 1.55[4] | 158,000 | 10 | 6 |
| 58EE[1] | 810[2]/855[4] | 432[2]/457[4] | 170 | 1.17 | 138,000 | 70/25 | 39/14 |
| 59FF[3] | 550[2]/600[4] | 288[2]/316[4] | 2000 | 13.79 | 210,000 | 50/0 | 28/0 |
| 60GG[3] | 550[2]/600[4] | 288[2]/316[4] | 2000 | 13.79 | 210,000 | 50/0 | 28/0 |
| 61HH[3] | 530[2]/600[4] | 277[2]/316[4] | 2200 | 15.17 | 210,000 | 60/0 | 33/0 |
| 62II[3] | 670[2]/700[4] | 354[2]/371[4] | 190 | 1.31 | 192,000 | 180/150 | 100/83 |
| 63JJ[3] | 600[2]/750[4] | 316[2]/399[4] | 500 | 3.45 | 235,000 | 180/30 | 100/17 |
| 64KK[3] | 600[2]/770[4] | 316[2]/410[4] | 525 | 3.62 | 283,000 | 170/0 | 94/0 |
| 65LL[3] | 775 | 413 | 550 | 3.79 | - | 0 | 0 |

Note: Numbers and letters in the first column (labeled "Point") refer to references and comments for Figure A-1.

[1]Catalytic reformer service.

[2]Average.

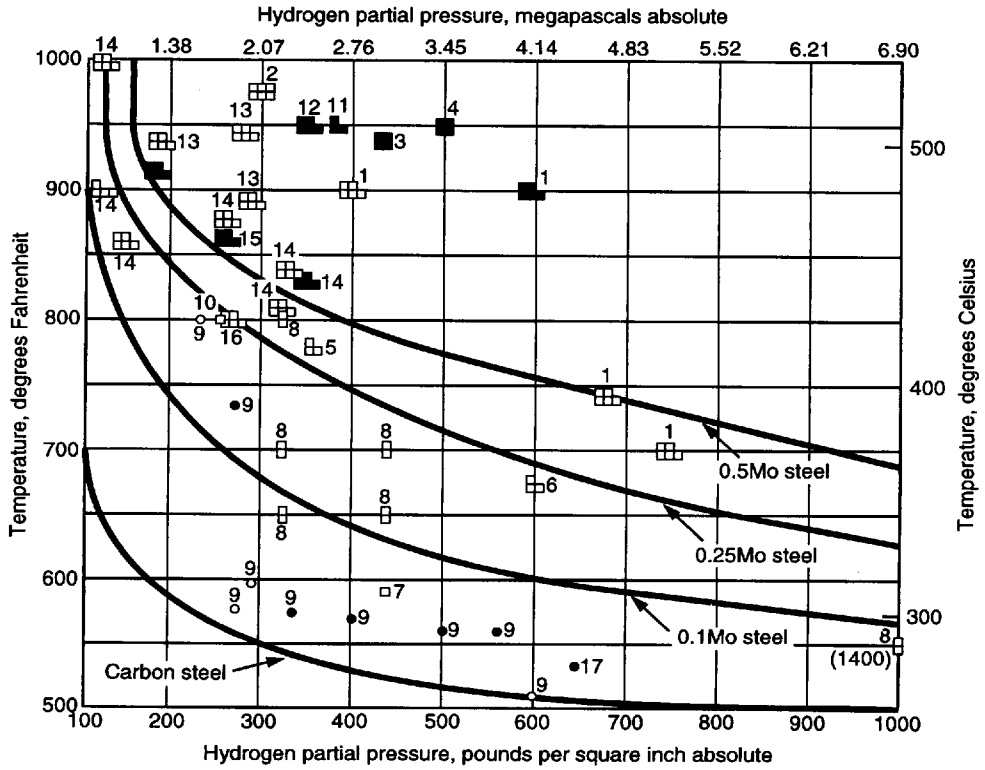
[3]Hydrodesulfurizer service.

[4]Maximum.

[5]Ammonia plant.

* API TG currently resolving these points.

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Notes
1. Mo has four times the resistance of Cr to HTHA.
2. Mo is equivalent to V, Ti, or Nb up to 0.1 percent
3. Si, Ni, and Cu do not increase resistance.
4. P, and S decrease resistance.

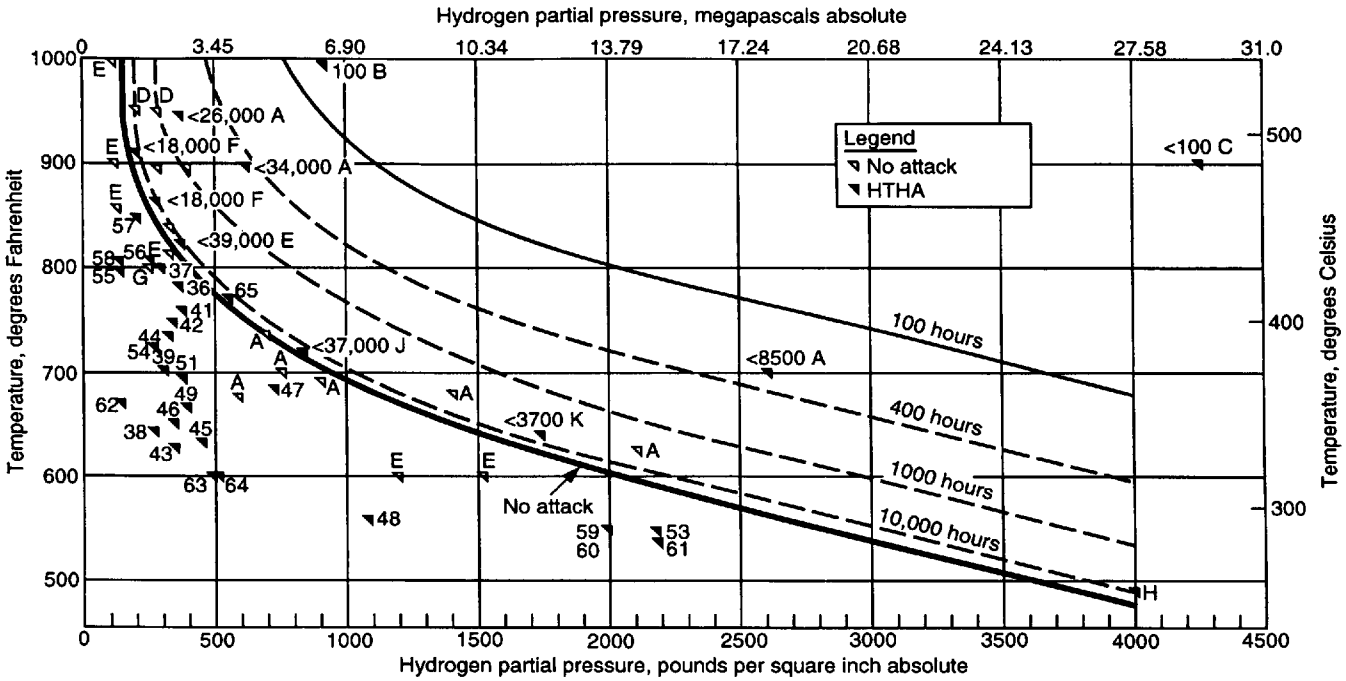
| Legend | | | | | | | |
|---|---|---|---|---|---|---|---|
| Equivalent molybdenum content | 0 | 0.01-0.10 | 0.11-0.20 | 0.21-0.30 | 0.31-0.40 | 0.41-0.50 | 0.51-0.60 |
| Satisfactory | ○ | □ | ▯ | ▯ | ⊞ | ⊞ | ⊞ |
| HTHA | ● | ■ | ▮ | ▮ | ■ | ■ | ■ |

| References | | Analysis Cr | Mo | V | Mo Equiv. |
|---|---|---|---|---|---|
| 1. | Shell Oil Company.* | | 0.50 | | 0.50 |
| 2. | Weld Deposits, D. J. Bergman.* | 0.79 | 0.39 | | 0.59 |
| 3. | Weld Deposits, D. J. Bergman.* | 0.80 | 0.15 | | 0.35 |
| 4. | Weld Deposits, D. J. Bergman.* | 0.50 | 0.25 | | 0.37 |
| 5. | Continental Oil Company.* | | 0.25 | | 0.25 |
| 6. | Standard Oil Co. of California.* | | 0.27 | | 0.27 |
| 7. | Standard Oil Co. of California.* | 0.05 | 0.06 | 0.08 | |
| 8. | A. O. Smith Corp.* | | | 0.13-0.18 | 0.11 |
| 9. | Shell Development Co., Drawing No. VT 659-2. | | | | |
| 10. | Amoco Oil Company.* | 0.04 | | | 0.01 |
| 11. | R. W. Manuel, Corrosion, 17(9), pp. 103-104, Sept. 1961 | 0.27 | 0.15 | | 0.22 |
| 12. | The Standard Oil Co. of Ohio.* | 0.11 | 0.43 | | 0.50 |
| 13. | Exxon Corporation.* | | | | |
| 14. | Union Oil of California.* | | | | |
| 15. | Amoco Oil Company.* | | | | |
| 16. | Standard Oil Co. of California. * | | | | |
| 17. | Gulf Oil Corporation.* | | | | |

*Private communication to Subcommittee on Corrosion (now Subcommittee on Corrosion and Materials).

Figure A-2—Steels in High Temperature Hydrogen Service Showing Effect of Molybdenum and Trace Alloying Elements

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Notes
1. This figure was adapted from Figure 3, Fourth Edition (1990) of this publication.
2. Numbered and lettered references for points in this figure refer to data listed in Table A-1 and Figure A-1 comments.

Figure A-3—Time For Incipient Attack of 0.5Mo Steels in High Temperature Hydrogen Service

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# APPENDIX B—HIGH TEMPERATURE HYDROGEN ATTACK OF 1.25 Cr-0.5Mo STEEL

The purpose of this appendix is to provide a brief summary of the information and experience regarding three case histories with high temperature hydrogen attack of 1.25Cr–0.5Mo steel.

The three recent experiences with high temperature hydrogen attack are listed in Table B-1, and the operating conditions are plotted in Figure B-1.

Table B-1—Experience with High Temperature Hydrogen Attack of 1.25Cr–0.5Mo Steel at Operating Conditions Below the Figure 1 Curve

| Case | Temperature | | Hydrogen Partial Pressure (Absolute) | | Service Years | Description |
|---|---|---|---|---|---|---|
| | °F | °C | psi | MPa | | |
| A | 960 | 516 | 331 | 2.28 | 26 | 1.5 NPS Schedule 80 nozzle was broken off a catalytic reformer outlet line during a shut down. Metallography indicated surface decarburization and intergranular cracking with bubbles. Cr content was 1.09 percent. |
| B | 977 | 525 | 354 | 2.44 | 14 | Blistering was detected with ultrasonic examination in catalytic reformer piping. Metallography indicated surface decarburization and blistering at non-metallic inclusions, with intergranular cracks growing from the blisters. Cr content was 1.10 percent. |
| C | 957/ 982 | 514/ 528 | 294/ 408 | 2.03/ 2.81 | 16 | Blistering near pipe inner surface. Examination showed decarburization between the inner surface and the blister. Gas analysis indicated methane in the blister. Cr content was 1.12 percent. |
| | Note 1 | Note 1 | Note 1 | Note 1 | | |

Note 1: Average conditions are reported as top number. Maximum condition reported as bottom number.

Cases A and B were reported by Chiyoda Corporation in Japan. Case C was originally reported by Merrick and Maguire of Exxon (see reference 10 in 2.2). The mechanisms of attack were similar in Cases B and C. That is, damage was in the form of internal blistering, with decarburization and intergranular cracking from the edges of the blisters. In Case A, however, attack resulted in intergranular separation. All three steels had chromium contents near 1.1 percent, near the 1.0 percent lower limit for 1.25Cr–0.5Mo steels. Additionally, the Case A steel had a relatively high impurity content with an $\bar{x}$ equal to 31.5, where $\bar{x}$ is defined as:

$$\bar{x} = \frac{10P + 5Sb + 4Sn + As}{100}, \text{ all elements in ppm.} \quad [37]$$



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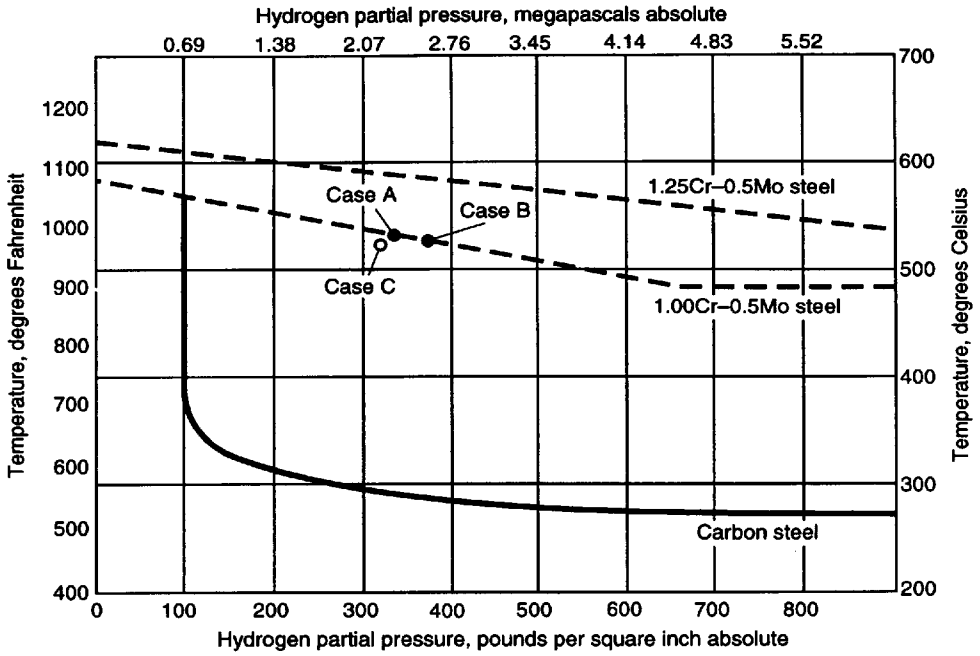



Figure B-1—Operating Conditions for 1.25Cr–0.5Mo Steels That Experienced HTHA Below the Figure 1 Curve

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# APPENDIX C—HIGH TEMPERATURE HYDROGEN ATTACK OF 2.25Cr-1Mo STEEL

The purpose of this appendix is to provide a brief summary of experience regarding a case history[a] with high temperature hydrogen attack of 2.25Cr–1Mo steel.

A recent experience with high temperature hydrogen attack is described in Table C-1, and the operating conditions are plotted in Figure C-1.

This case history may indicate that highly stressed components can suffer high temperature hydrogen attack at conditions below the curve in Figure 1. In this case history, the mixing tee was believed to be highly stressed by thermal stresses due to the mixing of hot and cooler hydrogen. Figure C-1 shows the operating conditions of both the hot upstream hydrogen and the mixed hydrogen downstream of the tee.

Table C-1—Experience With High Temperature Hydrogen Attack of 2.25Cr–1Mo Steel at Operating Conditions Below the Figure 1 Curve

| Temperature | | Hydrogen Partial Pressure (Absolute) | | Time in Service | Description |
|---|---|---|---|---|---|
| °F | °C | psi | MPa | Years | |
| 675/820 | 357/438 | 1385/1570 | 9.54/10.82 | >20 | A mixing tee for the hot and cold make-up hydrogen to a hydroprocessing unit leaked near the weld to the downstream piping. SEM examination indicated decarburization and fissuring along the internal surface of the tee. |
| See Note | See Note | See Note | See Note | | Although the leak path was not positively identified, it was concluded to be most likely due to fine, interconnected fissures. Some thermal fatigue cracking was also identified in the tee. Piping downstream of the tee was also found to have fissuring and internal decarburization to a depth of about 3.90 mils (0.1 mm) along the inside surface. The hot, upstream piping was not found to be attacked. |

Note: Average conditions are reported as first number. Maximum condition reported as second number.

[a]Communication to the API Subcommittee on Corrosion and Materials from Exxon Research and Engineering, August 1995.



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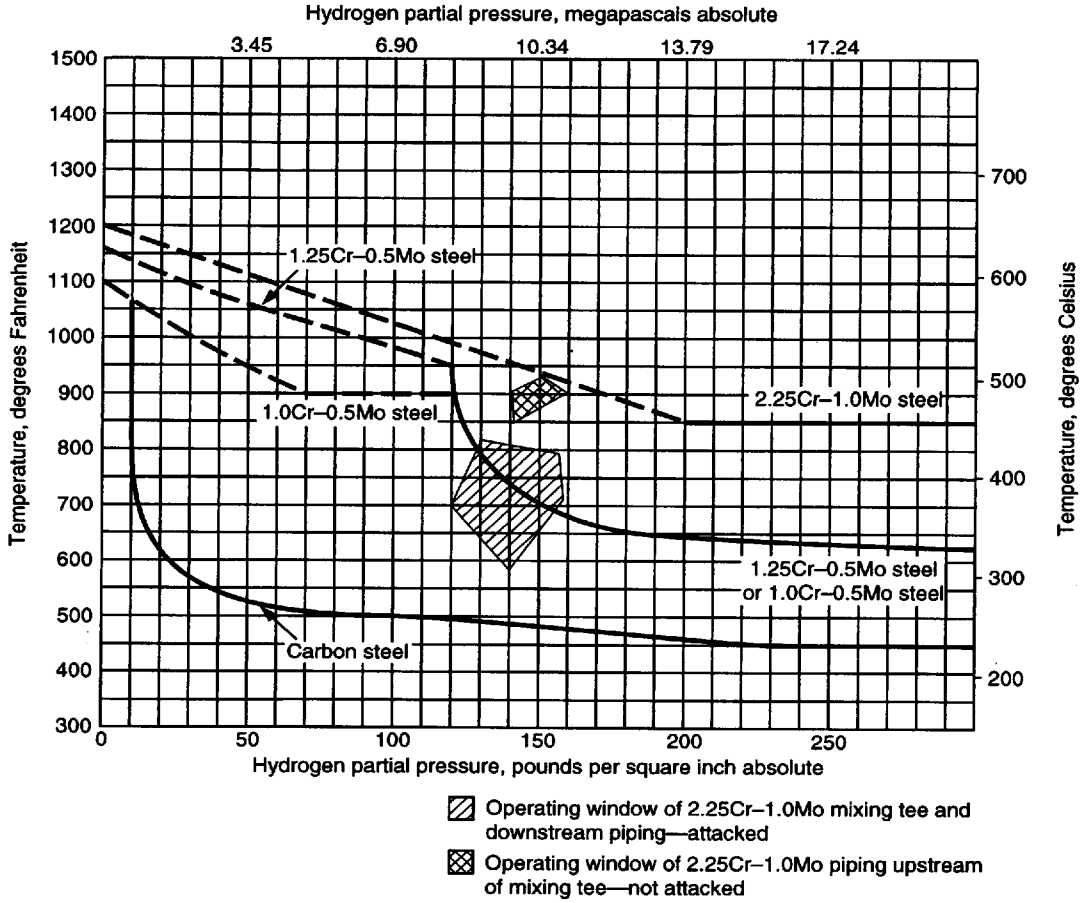



Figure C-1—Operating Conditions of 2.25Cr–1Mo Steels That Experienced HTHA Below the Figure 1 Curve

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# APPENDIX D—INSPECTION FOR HTHA TABLES

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## Table D-1—Summary of Inspection Methods for High Temperature Hydrogen Attack (HTHA)

| | Ultrasonic Methods | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Attenuation | | Backscatter | | | | | |
| | Velocity Ratio | Peak Amplitude | Spectral Analysis | Amplitude Based | Pattern Recognition | Spatial Averaging | Directional Dependence | Frequency Dependence | Conventional Shear Wave |
| Description | Ratio of shear and longitudinal velocity is measured. HTHA changes the ratio. | Dispersion of ultrasonic wave is measured by recording drop in amplitude of multiple echoes. HTHA increases attenuation. | The first backwall signal is analyzed in terms of amplitude versus frequency. HTHA will attenuate high frequency response more than low frequencies. | High frequency ultrasonic waves backscattered from within the metal are measured. HTHA causes a nonlinear increase in backscatter response. | High frequency ultrasonic waves backscattered from within the metal are analyzed. HTHA causes a rise and fall in backscatter pattern. | Backscatter data are collected over an area scanned. The signal is averaged to negate grain noise. | Compares backscatter signal as taken from ID and OD directions. HTHA damaged material will show a shift in indicated damage towards the exposed surface (ID). | Compares backscatter of two different frequency transducers. HTHA damaged material will show a shift in time. | Conventional shear wave UT routinely used for crack detection at welds. Higher frequencies increase detection capability. |
| Detection Capability | Has been shown to detect advanced stages of microscopic HTHA in base metal, away from welds. Can distinguish HTHA from plate laminations. | Has been shown to detect microscopic HTHA in base metal away from welds. | Has been shown to detect microscopic HTHA in base metal away from welds. | Has been shown to detect microscopic HTHA in base metal away from welds. | Has been shown to detect microscopic HTHA in base metal and weld metal. | Has been shown to detect microscopic HTHA in base metal and weld metal. | Has been shown to detect microscopic HTHA in base metal and weld metal. | Has been shown to detect microscopic HTHA in base metal and weld metal. | Can reliably detect HTHA only after cracks have formed. Cannot detect microscopic HTHA. |
| Advantages | Not affected by inclusions. No prior inspection history needed for interpretation. | Simple to use. | Very sensitive to internal microfissuring due to HTHA. Can be used to distinguish laminations from HTHA. | Very sensitive to internal microfissuring due to HTHA. Can be used for scanning. Can give an indication of depth of HTHA. Can be automated in either a B-scan or C-scan mode. Can be used as a monitoring tool. | Very sensitive to internal microfissuring due to HTHA. Can be used for scanning. Can give an indication of depth of HTHA. Can discriminate HTHA from inclusions. | Sensitive to internal microfissuring due to HTHA. Can be used to improve detection of initial stages of HTHA and to determine depth of damage. | Very sensitive to internal microfissuring due to HTHA. Can be used to distinguish HTHA from other internal defects such as laminations. | Very sensitive to internal microfissuring due to HTHA. Can be used to distinguish HTHA from other internal defects such as laminations. | Can scan full coverage of welds from the OD. |
| Limitations | Covers only local spot where probe is held. Cannot be used for scanning large areas. Cannot detect early stages of HTHA. Typically only detects damage greater than 20 percent through wall. Cladding can cause false interpretation. | Covers only local spot where probe is held. Cannot be used for scanning. ID or OD surface corrosion can give false readings. Needs parallel surfaces. Thick materials decrease sensitivity. Difficult to get similar repeat readings when used as a monitor- | Technique is best when used as a comparison of a clean non-HTHA area versus a suspect area. | Inclusions, large grains, ID pitting, laminar defects, or scale can give false indications of HTHA. Damage from HTHA attenuates backscatter signal. | Inclusions, large grains, ID pitting, laminar defects, or scale can give false indications of HTHA. Damage from HTHA attenuates backscatter signal. | Not a primary method, usually a complementary method. | Requires access to both ID and OD surfaces. Does not work well on clad equipment. | Not a primary method, usually a complementary method. Does not work well on very shallow HTHA damage. | Cannot detect the microscopic stages of HTHA. Detects only the advanced stages after cracks have formed. |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | laminations from HTHA. | Can give an indication of depth of HTHA. Can be automated in either a B-scan or C-scan mode. Can be used as a monitoring tool. | Can give an indication of depth of HTHA. Can discriminate HTHA from inclusions. | of initial stages of HTHA and to determine depth of damage. | HTHA from other internal defects such as laminations. | HTHA from other internal defects such as laminations. | |
| Limitations | Covers only local spot where probe is held. Cannot be used for scanning large areas. Cannot detect early stages of HTHA. Typically only detects damage greater than 20 percent through wall. Cladding can cause false interpretation. | Covers only local spot where probe is held. Cannot be used for scanning. ID or OD surface corrosion can give false readings. Needs parallel surfaces. Thick materials decrease sensitivity. Difficult to get similar repeat readings when used as a monitoring program. | Technique is best when used as a comparison of a clean non-HTHA area versus a suspect area. | Inclusions, large grains, ID pitting, laminar defects, or scale can give false indications of HTHA. Damage from HTHA attenuates backscatter signal. | Inclusions, large grains, ID pitting, laminar defects, or scale can give false indications of HTHA. Damage from HTHA attenuates backscatter signal. | Not a primary method, usually a complementary method. | Requires access to both ID and OD surfaces. Does not work well on clad equipment. | Not a primary method, usually a complementary method. Does not work well on very shallow HTHA damage. | Cannot detect the microscopic stages of HTHA. Detects only the advanced stages after cracks have formed. |
| Recommendations | Recommended for base metal HTHA detection when advanced HTHA is indicated by other methods. | Not currently recommended for HTHA inspection. | Used as a complementary technique after backscatter method indicates possible damage. | Recommended only when used with other techniques. Should be limited to the initial screening. | Recommended only when used with other techniques as the first step of inspection. | Used as a complementary technique when depth of damage cannot be clearly identified. | Used as a complementary technique after amplitude based backscatter method indicates possible damage. | Used as a complementary technique after amplitude based backscatter method indicates possible damage. | Recommended for external inspection but recognize its limitation for advanced stages of attack and inability to find microscopic stages of attack. |

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## Table D-2—Summary of Inspection Methods for High Temperature Hydrogen Attack (HTHA)

| | Other Methods | | | | |
|---|---|---|---|---|---|
| | Magnetic Particle | Field Metallography and Replication | Radiography | Visual | Acoustic Emission |
| Description | Conventional wet fluorescent AC yoke magnetic particle inspection used for detection of cracks at a surface. Blending the welds and sanding smooth increases sensitivity. | Polish and etch as in a creep evaluation looking for microscopic damage. Replicas can be taken for laboratory analysis. | Conventional radiography used to inspect welds for cracks. | Internal visual inspection of pressure vessels for surface blistering. | Monitors the sound that cracks emit when they are stressed. |
| Detection Capability | Can detect HTHA only after cracks have formed. Cannot detect microscopic damage. | Can detect and distinguish microfissuring and micro voids due to HTHA from other indications very reliably. | Can detect HTHA only after macro size cracks have formed. Cannot detect microscopic damage. | Blisters are readily apparent when present. However, HTHA may frequently occur without the formation of surface blisters. | Reported to be capable of detecting cracks. Currently not known whether microscopic damage can be detected. |
| Advantages | Crack indications can be seen visually and little interpretation is required. | Only nondestructive confirmation method. Can be used at welds and base metal. | Radiographic film gives a record of detected cracks. Additionally, radiography can sometimes be used for crack detection without insulation removal, although sensitivity with insulation in place may be poor. | No special inspection tools are needed. Blister interpretation is clear. | Capability for monitoring a large system including piping and pressure vessels. Potentially offers a technique for identifying areas needing follow-up inspection. May offer a method for full coverage of base metal. |
| Limitations | Cannot detect the microscopic stages of HTHA. Detects only the advanced stages after cracks have already formed. Only detects surface cracks. Exam is performed from the ID. Cannot determine the depth of HTHA damage. | Limited to ID surface. Cladding must be removed if present. Best if 1/16 to 1/8 inch (2 to 3 mm) of material is removed to reveal subsurface damage. Cannot nondestructively determine the depth of HTHA damage. | Cannot detect the microscopic stages of HTHA. May detect the advanced stages after cracks have already formed. | HTHA frequently occurs without the formation of surface blisters. Blisters, when present, are likely to be an indication of advanced HTHA. Cracking is not evident visually. | Not a proven technique for HTHA detection. Needs an applied stress during the test, usually by hydrostatic testing. Another test method uses thermal stress during equipment cooldown. |
| Recommendations | Recommended for internal inspection of pressure vessels to use in addition to UT techniques, recognizing its limitation for advanced stages of HTHA and not finding microscopic stages of HTHA. | Can be used to follow-up indications from other methods or at suspected hot spots where damage is suspected. | Not recommended for general HTHA detection. May be useful for verification of shear wave UT indications. | Recommended for internal inspection of pressure vessels to use in addition to UT & MT techniques. | Additional development work and field trials recommended. Not currently recommended as a primary method for HTHA detection. |

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# APPENDIX E—REQUEST FOR NEW INFORMATION

The API Subcommittee on Corrosion and Materials collects data on the alloys shown in all figures or similar alloys that may come into use. Revisions to the curves will be published as the need arises.

For the existing curves, data are desired for instances of high temperature hydrogen attack (HTHA) damage that occur above or below the curve for the steel involved; data are also desired for successful experience in the area above the curve for the steel involved. For chromium-molybdenum steels not included on the existing figures, data for successes and HTHA damage in any meaningful area are desired.

The following data sheet is provided for the reader's convenience in submitting new data. Available data should be furnished by inserting information in the spaces provided and checking the appropriate answer where a selection is indicated. Any additional information should be attached.

While both hydrogen partial pressure and temperature are important, particular attention should be given to obtaining the best estimate of accurate metal temperature. One method of obtaining more accurate data for a specific area is to attach a skin thermocouple to the area that previously exhibited high temperature hydrogen damage.

The completed form should be returned to the following address:

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STD.API/PETRO RP 941-ENGL 1997 ■ 0732290 0563361 898 ■



# Data Sheet for Reporting High Temperature Hydrogen Attack of Carbon and Low-Alloy Steels

Date _____ File No. _____

By _____
(Name, Company, Address)

1. (a) ASTM specification (or equivalent) for the steel: _____

   (b) Design Code _____

2. (a) Composition of steel (wt %) Fe_____ Cr_____ Mo_____ V_____ Ni_____ P_____ Sn_____
   Ti_____ Nb_____ C_____ Si_____ Mn_____ S_____ As_____

   (b) Steel protection: None_____ Weld overlay material _____ Sb _____
   Cladding material_____ Other_____

   (c) Thickness Base metals _____ Weld overlay or cladding (if any) _____

3. Heat treatment: Port weld heat treatment Yes_____ No_____ Temperature/Time _____ °F/hrs.
   Normalized and tempered Yes_____ No_____ Tempering Temperature _____ °F
   Quenched and tempered Yes_____ No_____ Tempering Temperature _____ °F
   Other _____

4. Physical properties (prior to exposure): Yield strength (actual) _____ pounds per square inch
   Ultimate strength (actual) _____ pounds per square inch

5. Temperature: Process: Average_____ °F Maximum_____ °F
   Metal: Average_____ °F Maximum_____ °F

6. Hydrogen partial pressure: _____ pounds per square inch absolute Hydrogen purity_____ %

7. Calculated operating stress: _____ pounds per square inch absolute

8. Microhardness: For a failure, at or near crack:_____
   For successes: Weld: _____ Base material _____
   Heat-affected zone: _____

9. Days in service: Total_____ At maximum temperature _____

10. Type of failure
    Surface decarburization Yes_____ No_____ Surface cracking Yes_____ No_____
    Internal decarburization Yes_____ No_____ Internal fissuring Yes_____ No_____
    Blisters Yes_____ No_____

11. Location of failure (include photograph):
    Weld metal Yes_____ No_____ Heat-affected zone Yes_____ No_____
    Base material Yes_____ No_____
    Other _____

12. The type of process unit involved.

13. Type of equipment (piping, vessels, heat exchanger, etc.)

14. Submit a photomicrograph showing typical failure and grain structure. Include 100x and 500x photomicrographs, plus any other appropriate magnifications. Attach any reports, if available. Please note any unusual circumstances.

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